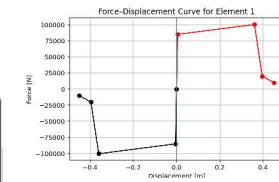
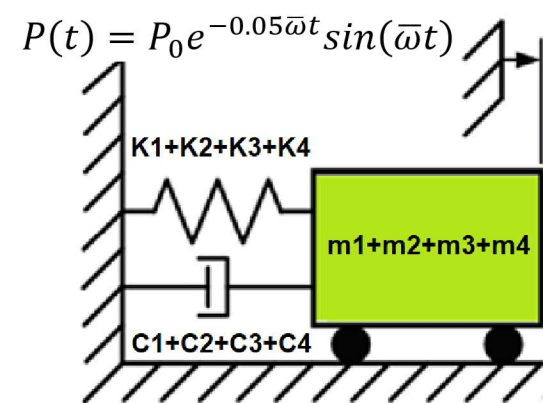


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

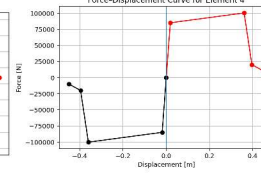
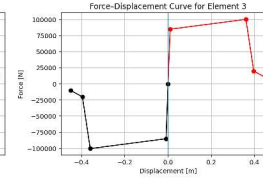
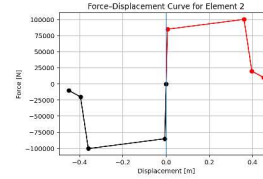
SENSITIVITY ANALYSIS OF SINGLE-DEGREE-FREEDOM (SDOF) STRUCTURES USING FREE-VIBRATION EFFECTS OF INITIAL DISPLACEMENT, STRUCTURAL ELASTIC STIFFNESS, STRUCTURAL DUCTILITY RATIO AND OVER-STRENGTH FACTOR ON OUTPUT KEY PARAMETERS SUCH AS EQUIVALENT VISCOUS DAMPING RATIO FROM NONLINEAR DYNAMIC ANALYSES USING PYTHON AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)

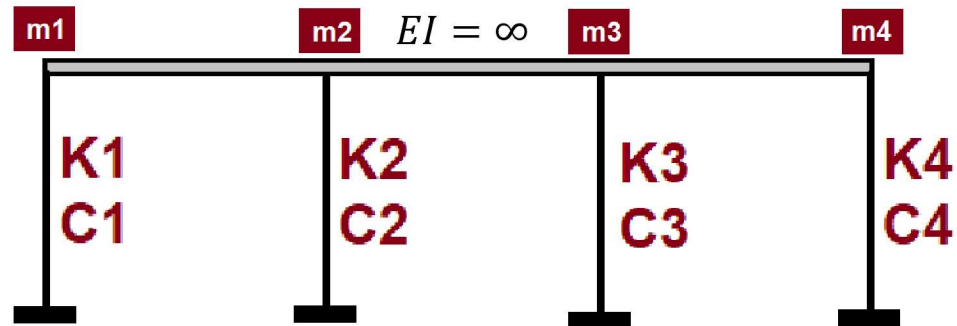
INCREMENTAL
DISPLACEMENT



Initial Displacement
Initial Velocity
Initial Acceleration



INCREMENTAL
DISPLACEMENT



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

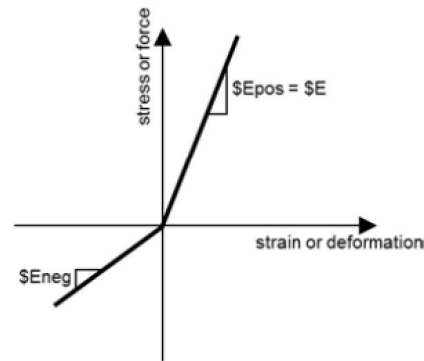
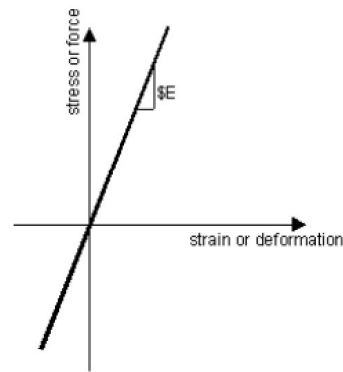
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

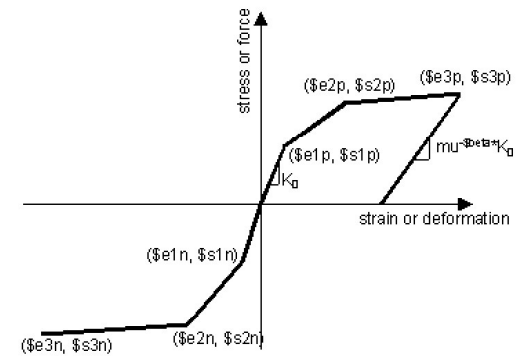
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL

Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

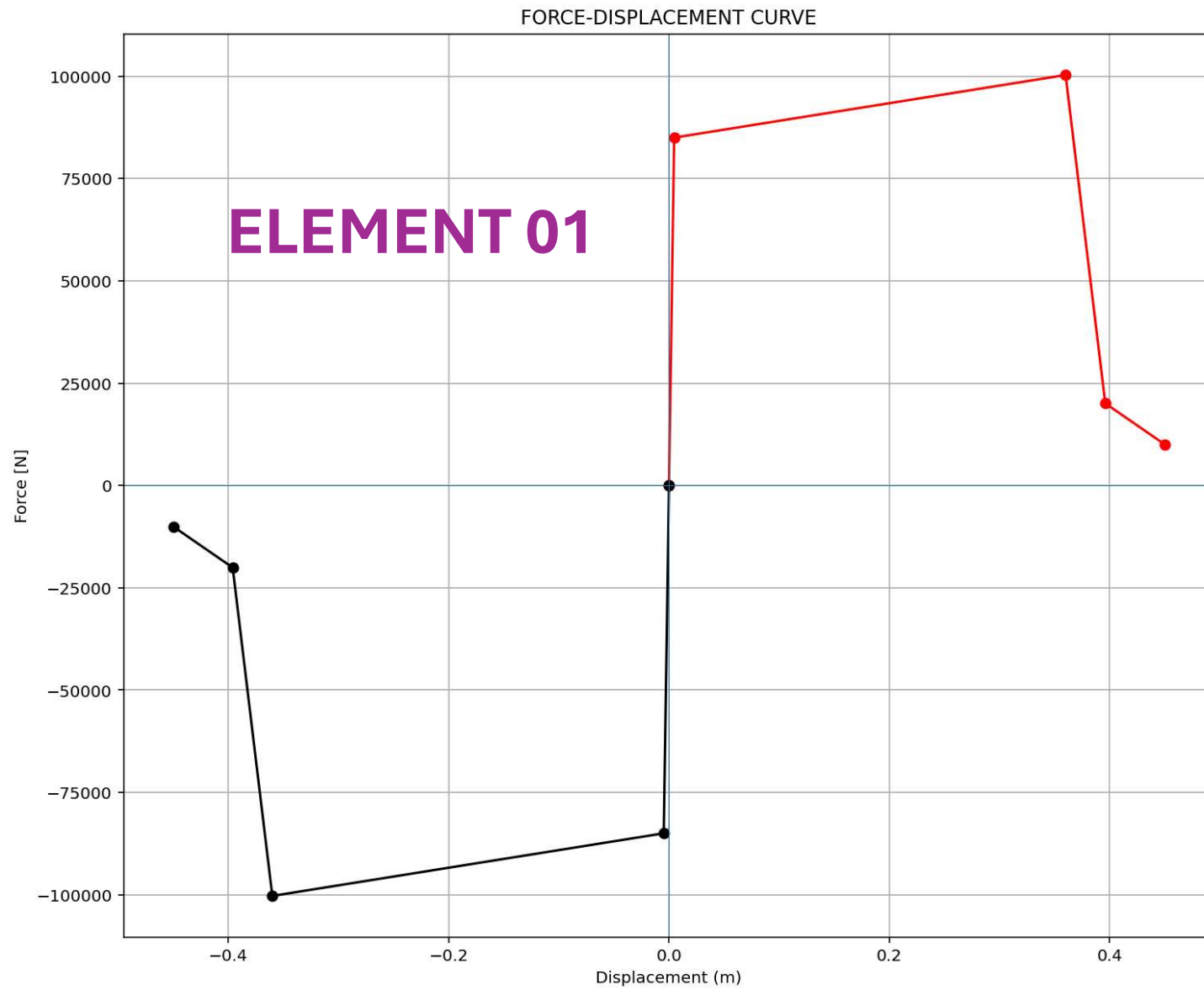
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

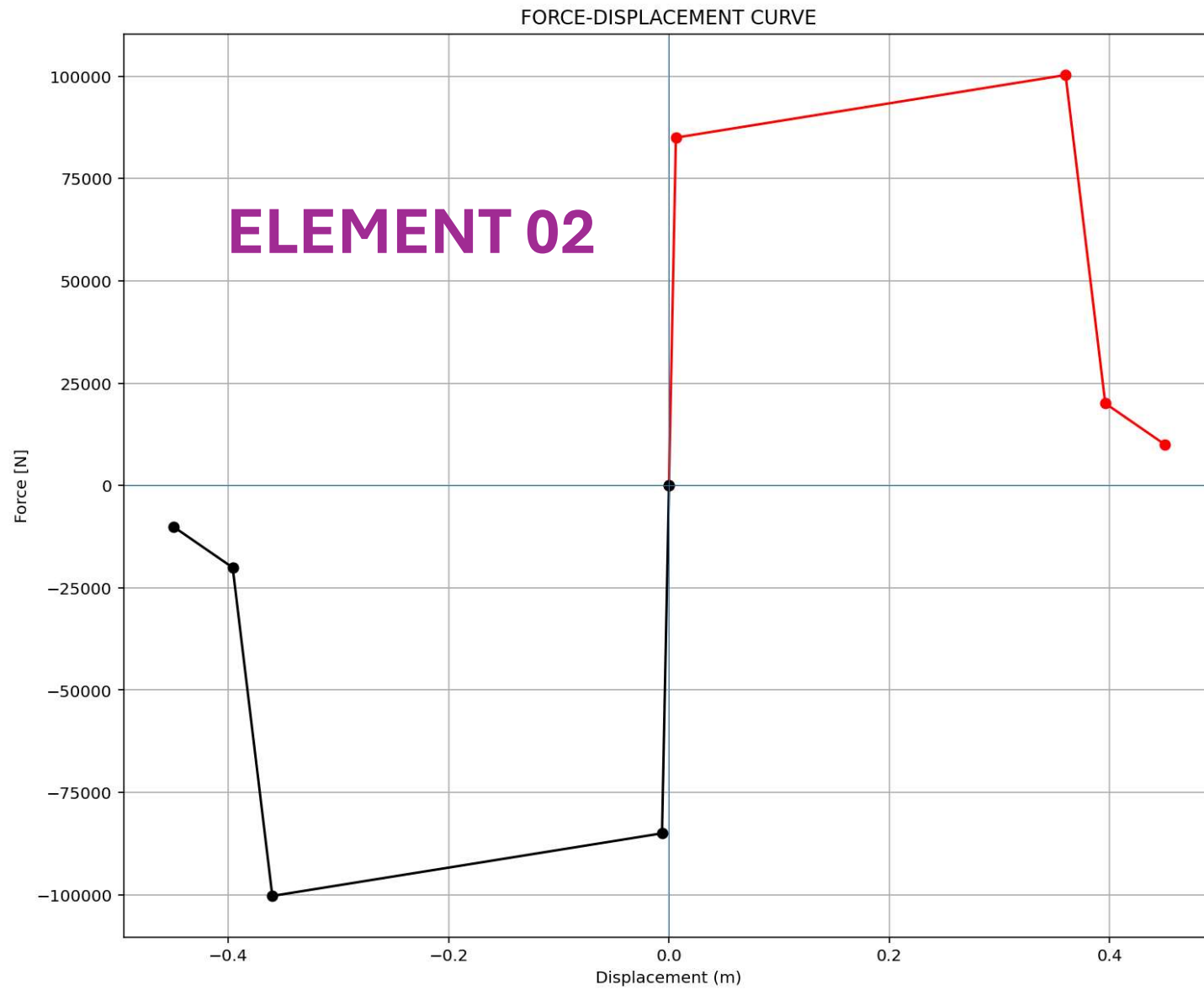
where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{\text{sec}} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

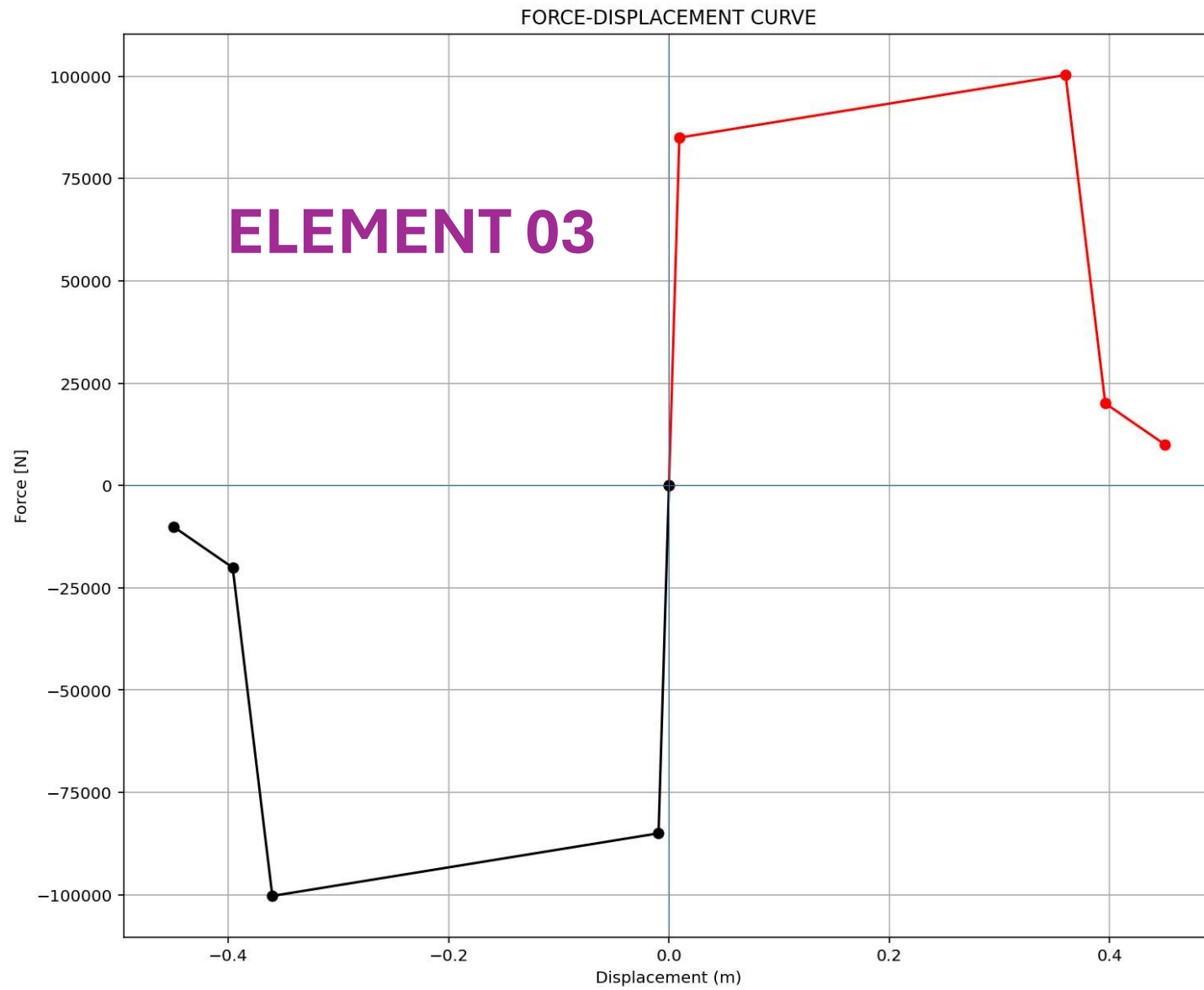
The equivalent viscous damping ratio is then:

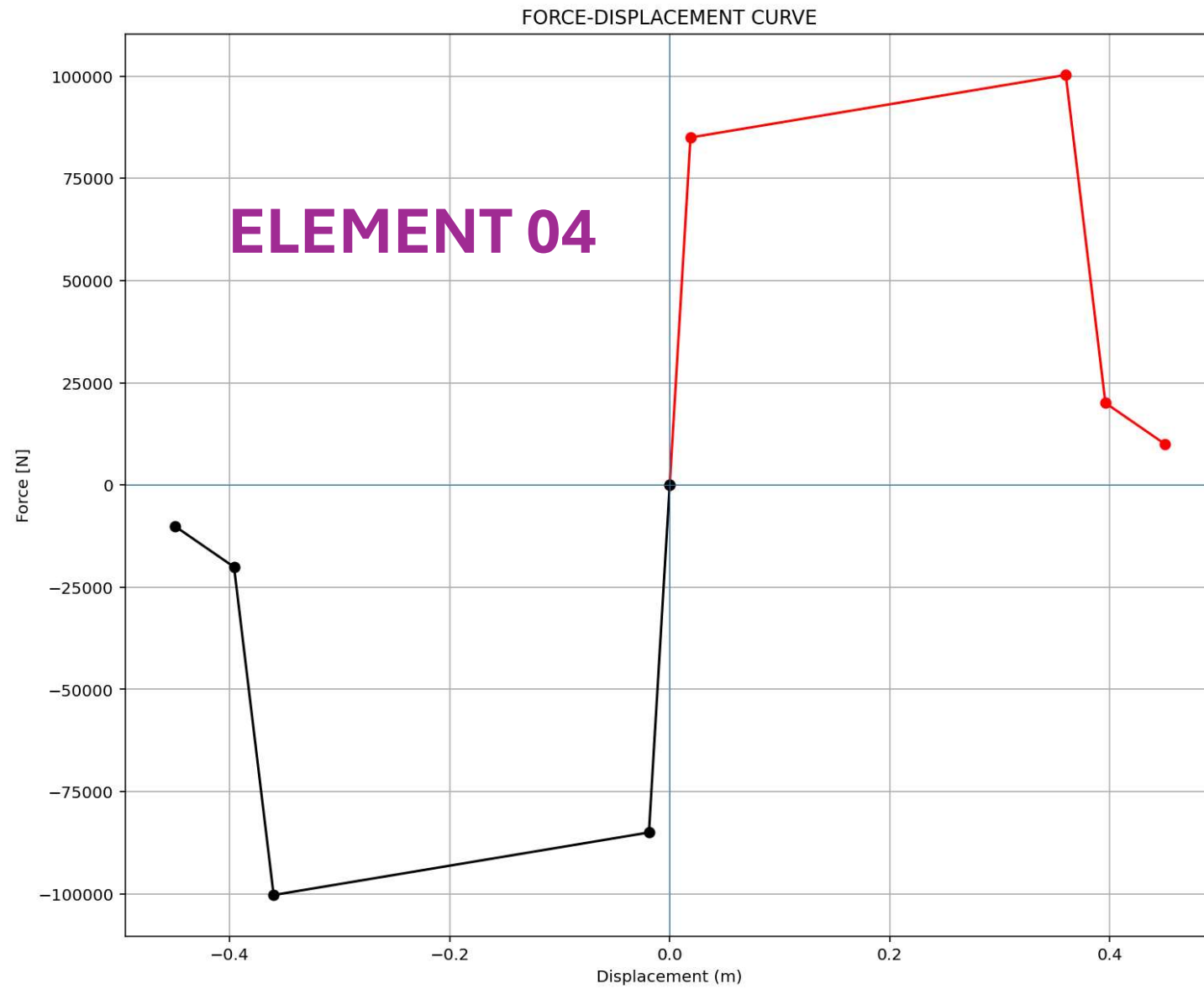
$$\xi_{eq} = \frac{100 E_d}{4\pi E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.









FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

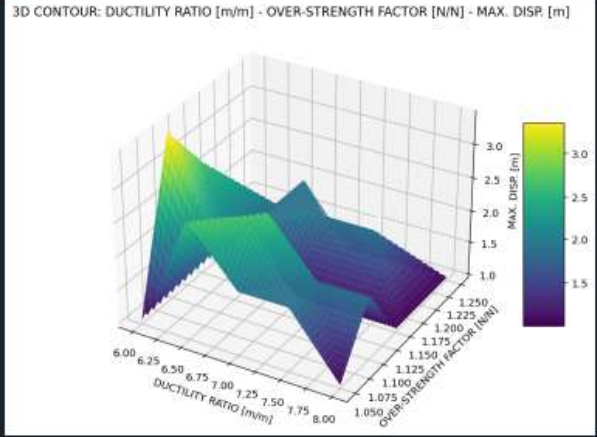
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...(t)_SENSITIVITY_ANALYSIS_SDOF_FOUR_ELEMENTS_FV.py

P(t)_SENSITIVITY_A...OUR_ELEMENTS_FV.py x PLOT_CONTOUR_3D_2D_FUN.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # SENSITIVITY ANALYSIS OF SINGLE-DEGREE-FREEDOM (SDOF) STRUCTURES USING FREE-VIBRA
4 # EFFECTS OF INITIAL DISPLACEMENT, STRUCTURAL ELASTIC STIFFNESS, STRUCTURAL DUCTILITY
5 #AND OVER-STRENGTH FACTOR ON OUTPUT KEY PARAMETERS FROM NONLINEAR DYNAMIC ANALYSES USING PYTHON
6 #
7 #-----
8 # FREE VIBRATION ANALYSIS USING INITIAL DISPLACEMENT
9 #-----
10 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
11 # EMAIL: salar.d.ghashghaei@gmail.com
12 #####
13 1. This script performs a "sensitivity analysis of a single-degree-of-freedom (SDOF) structure"
14 2. The objective is to study the effects of "initial displacement, structural elastic stiffness,
15 3. Structural properties such as "yield force, ultimate force, elastic stiffness, hardening rat
16 4. Elastic and plastic periods of the system are computed and reported.
17 5. Key seismic performance parameters are calculated, including "over-strength ( $\Omega_0$ )", "ductilit
18 6. A nonlinear "hysteretic spring model" is used to represent structural behavior with strength
19 7. A "viscous damper" is added to simulate energy dissipation.
20 8. The analysis considers "free vibration due to initial displacement" (no external ground moti
21 9. The function `ANALYSIS_SDOF` builds the OpenSees model, applies initial conditions, and runs
22 10. "Newmark time integration" and "Newton-Raphson iteration" are used for nonlinear solution.
23 11. "Rayleigh damping coefficients" are computed from the first eigenvalue.
24 12. Time histories of "displacement, velocity, acceleration, base reaction, stiffness, and damc
25 13. A "ductility-based damage index" is calculated at each time step.
26 14. The effective stiffness degradation is monitored during the response.
27 15. Multiple simulations are executed by looping over ranges of "mass, initial displacement, du
28 16. For each simulation, "maximum response values" are extracted.
29 17. Damage index values are limited between "0% and 100%".
30 18. All results are stored for post-processing and statistical analysis.
31 19. The script reports completion of each simulation and total runtime.
32 20. Overall, the code provides a "parametric nonlinear dynamic assessment" of SDOF systems unde
33
34 # BOOK: Dynamics of Structures in SI Units - Anil Kumar Chopra
```

34 %

3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]



IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 996, Col 1 UTF-8 CRLF RW Mem 78%

Spyder (Python 3.12)

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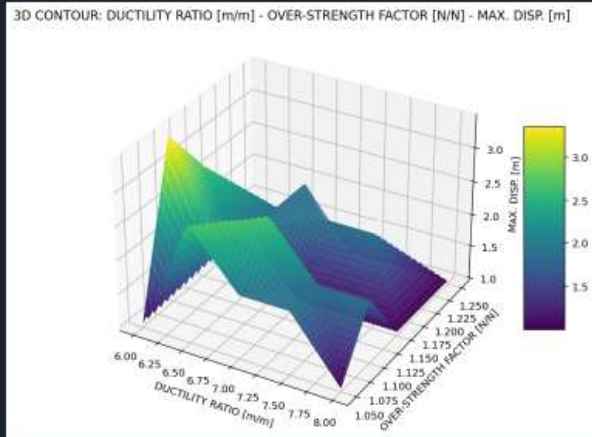
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P(t)_SENSITIVITY_A...OUR_ELEMENTS_FV.py x PLOT_CONTOUR_3D_2D_FUN.py x

```
1 def PLOT_CONTOUR_3D_2D_FUN(TAG, X, Y, Z, XLABEL, YLABEL, ZLABEL):
2     # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
3     import numpy as np
4     import matplotlib.pyplot as plt
5     from scipy.interpolate import griddata
6     # Convert to NumPy arrays
7     X = np.array(X)
8     Y = np.array(Y)
9     Z = np.array(Z)
10
11     # Create grid for contour plot
12     xi = np.linspace(min(X), max(X), 100)
13     yi = np.linspace(min(Y), max(Y), 100)
14     xi, yi = np.meshgrid(xi, yi)
15
16     # Interpolate Z values on the grid
17     #zi = griddata((X, Y), Z, (xi, yi), method='nearest')
18     zi = griddata((X, Y), Z, (xi, yi), method='linear')
19     #zi = griddata((X, Y), Z, (xi, yi), method='cubic')
20
21     # Plot 3D contour
22     fig = plt.figure(TAG, figsize=(10, 7))
23     ax = fig.add_subplot(111, projection='3d')
24     contour = ax.plot_surface(xi, yi, zi, cmap='viridis', edgecolor='none')
25
26     ax.set_xlabel(XLABEL)
27     ax.set_ylabel(YLABEL)
28     ax.set_zlabel(ZLABEL)
29
30     fig.colorbar(contour, ax=ax, shrink=0.5, aspect=5)
31     plt.title(f'3D CONTOUR: {XLABEL} - {YLABEL} - {ZLABEL}')
32     plt.show()
33
34
35 fig, ax = plt.subplots(figsize=(10, 7))
```

34 %

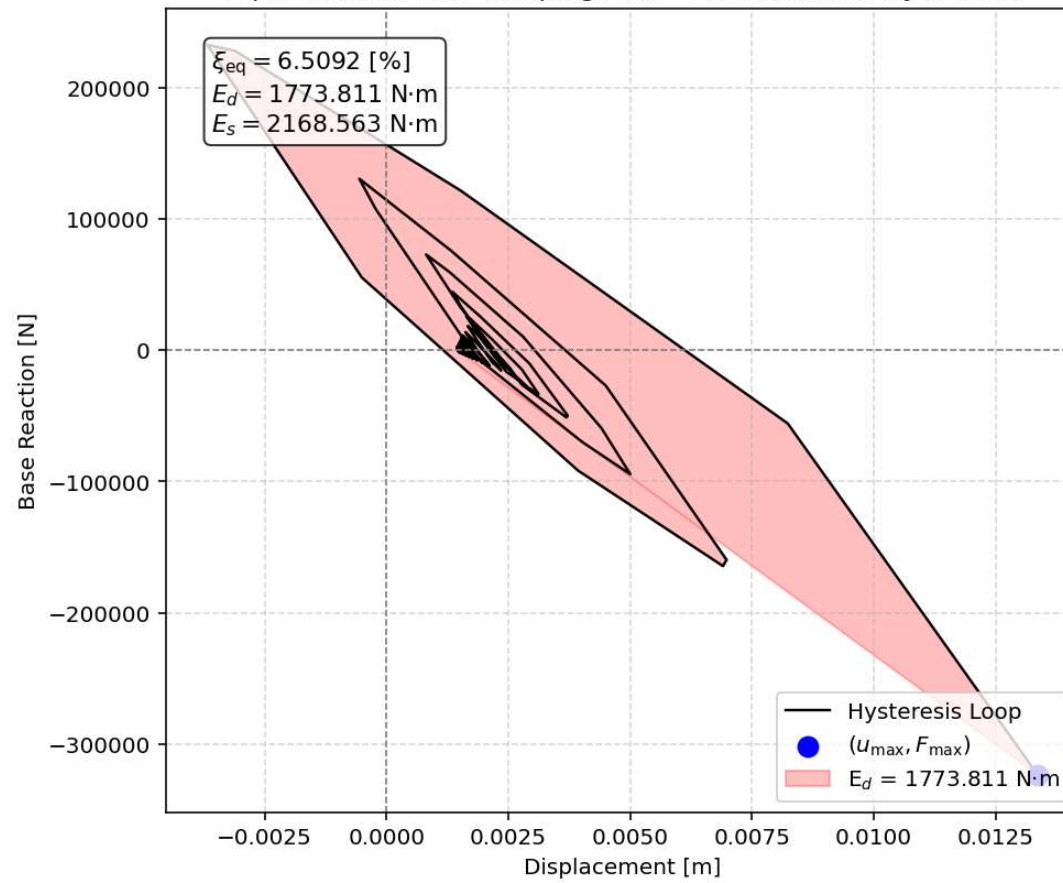
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]



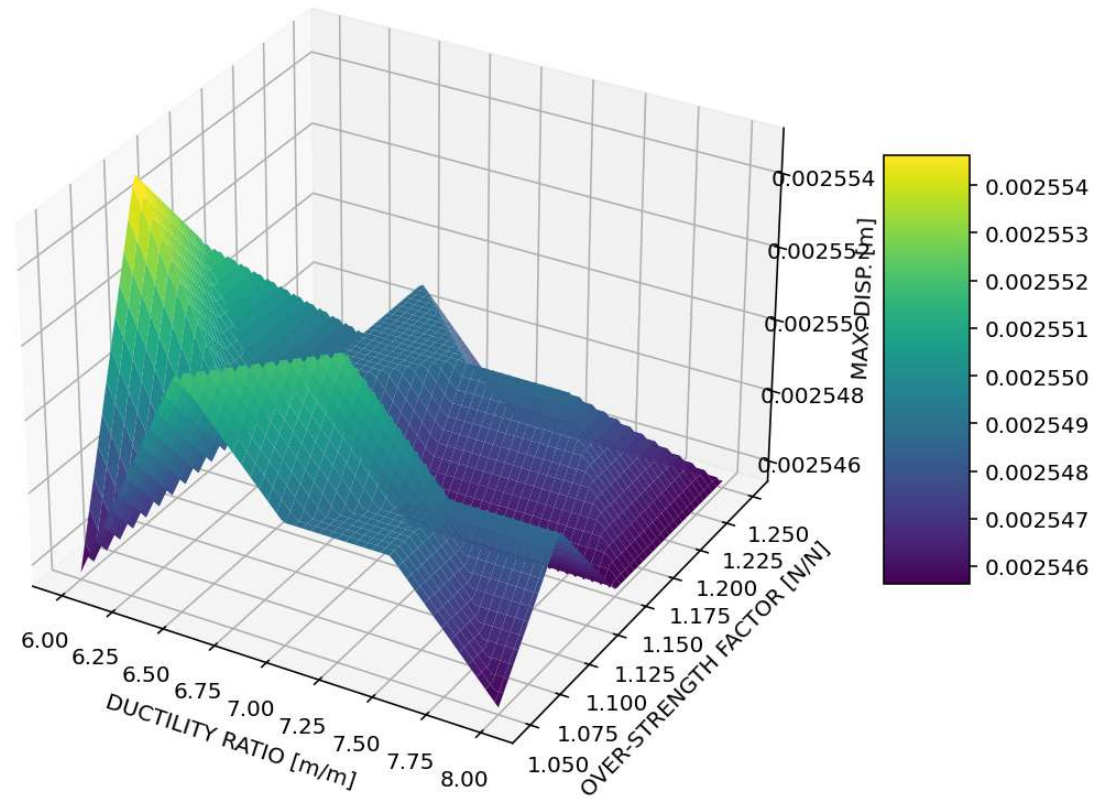
IPython Console Files Help Variable Explorer Debugger Plots History

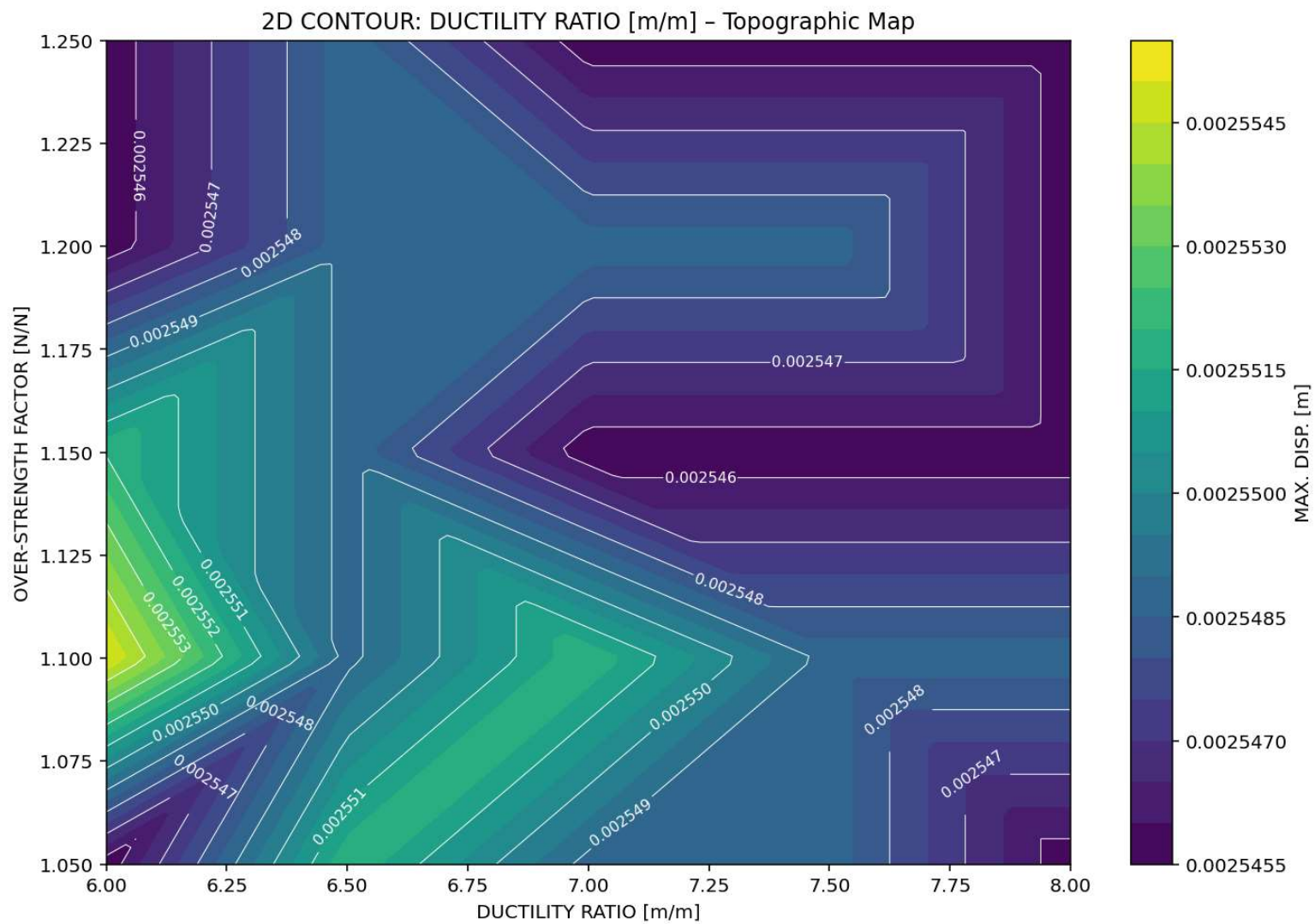
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 78%

Equivalent viscous damping ratio - Free-vibration Hysteresis

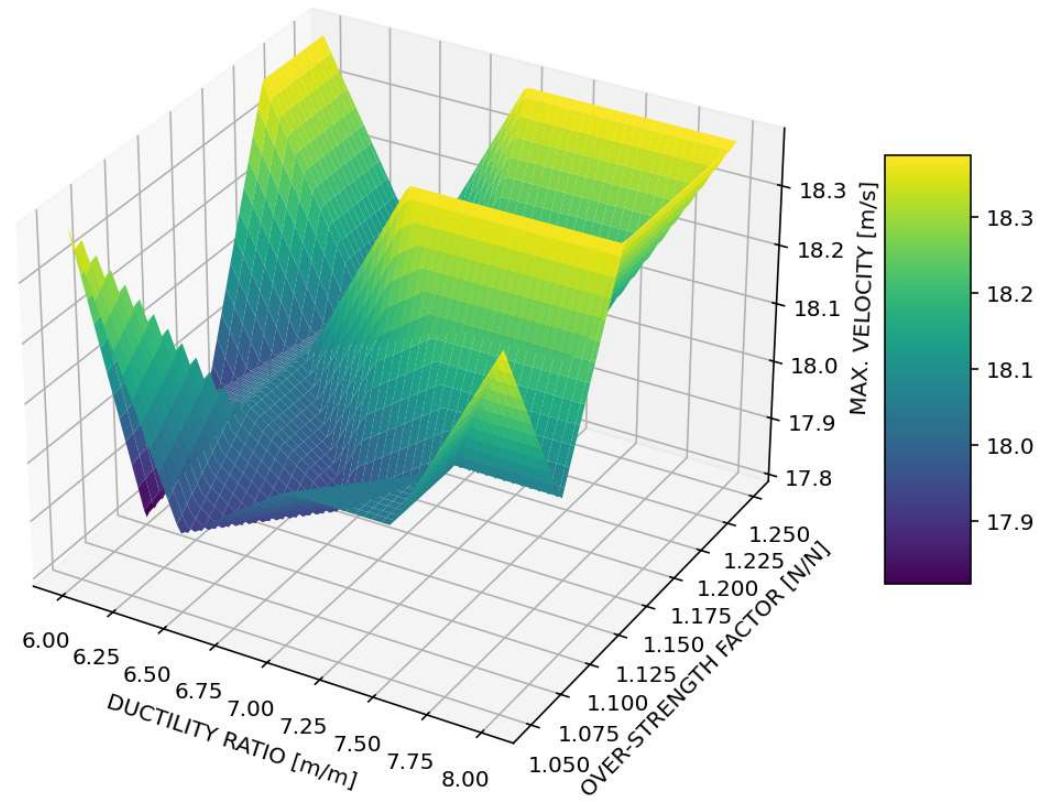


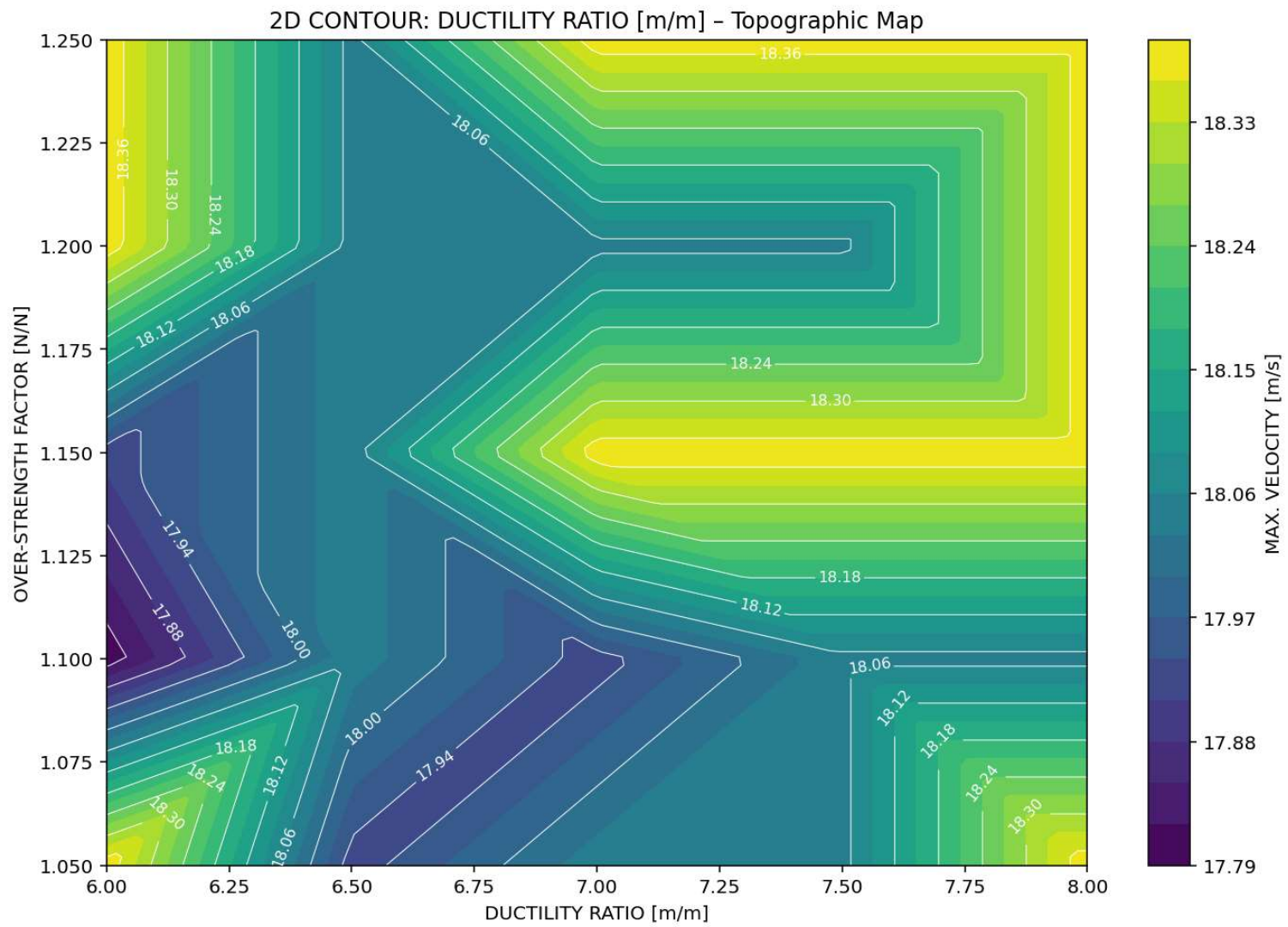
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]



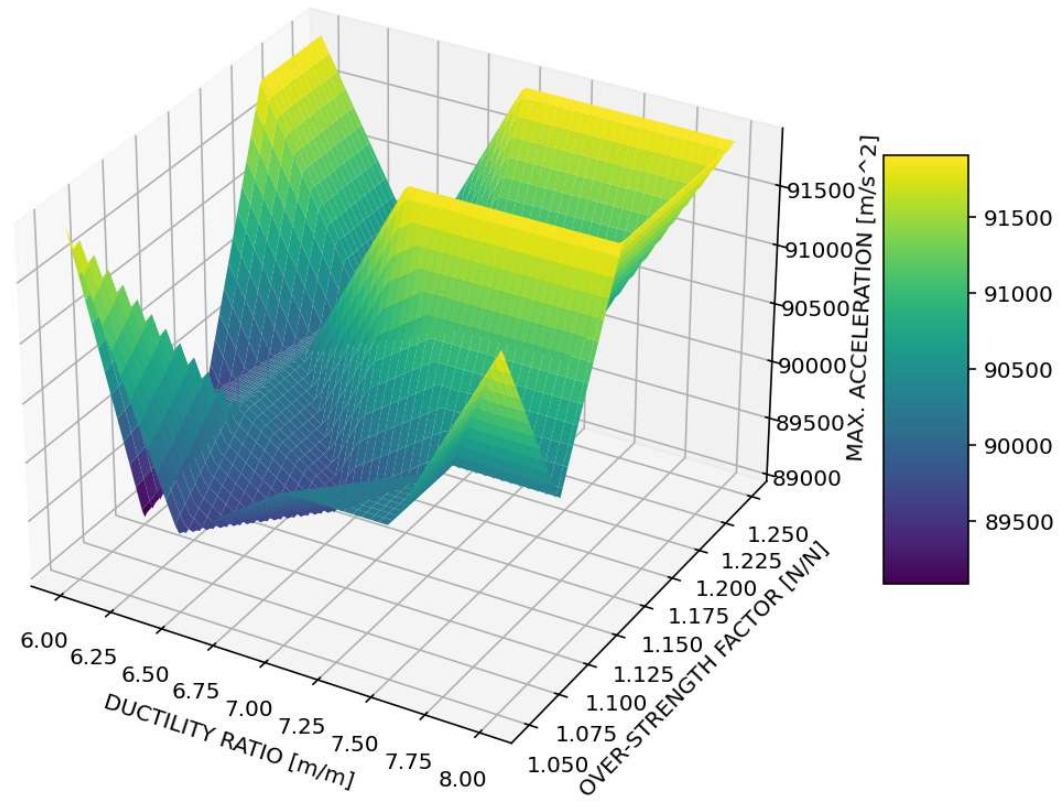


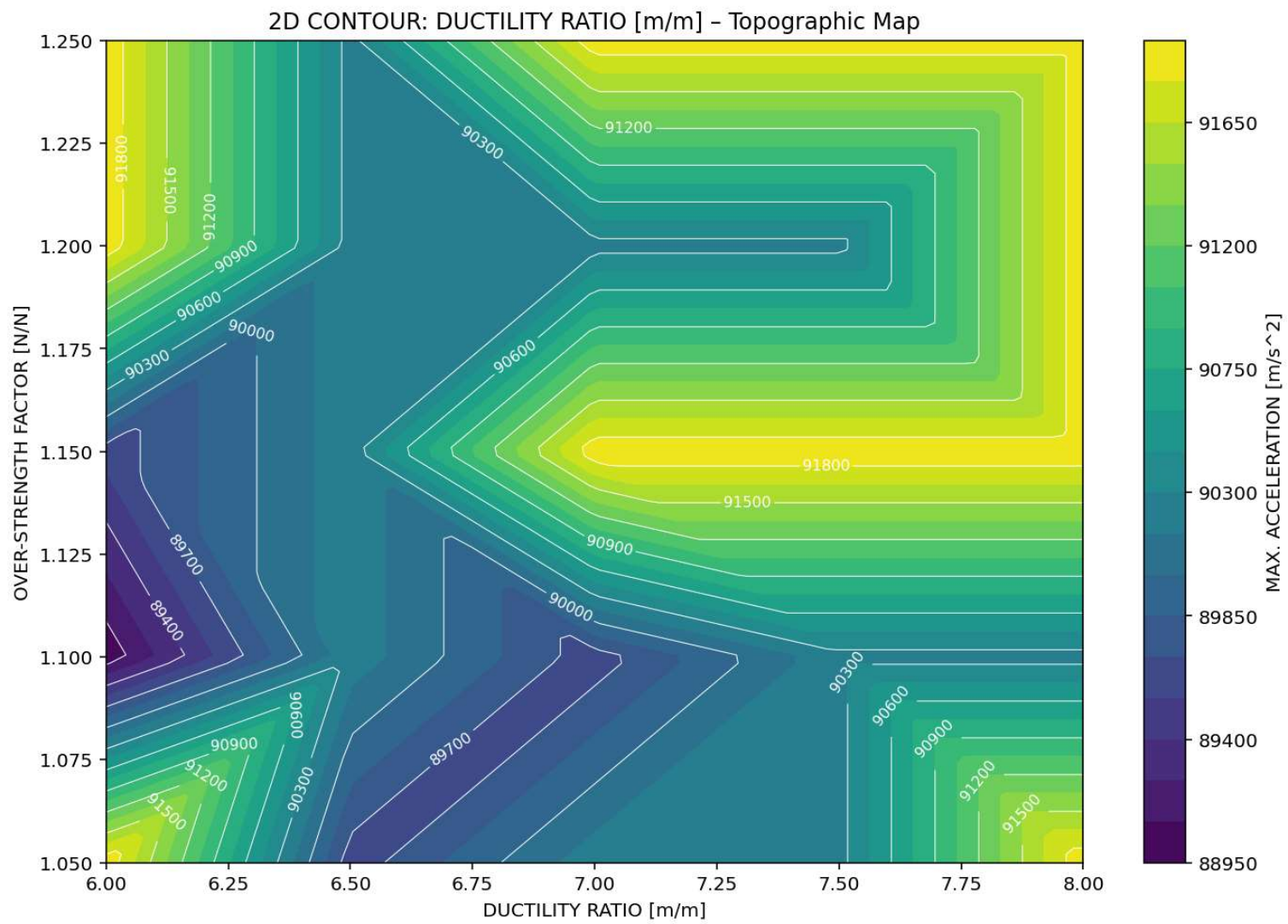
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. VELOCITY [m/s]



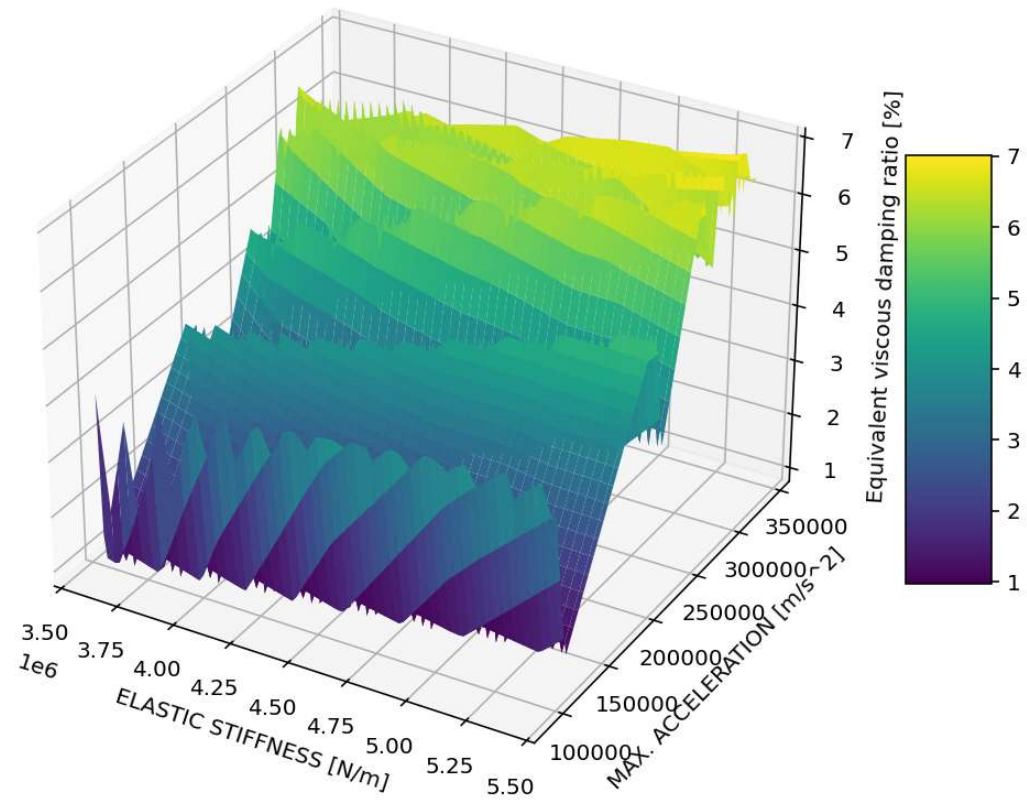


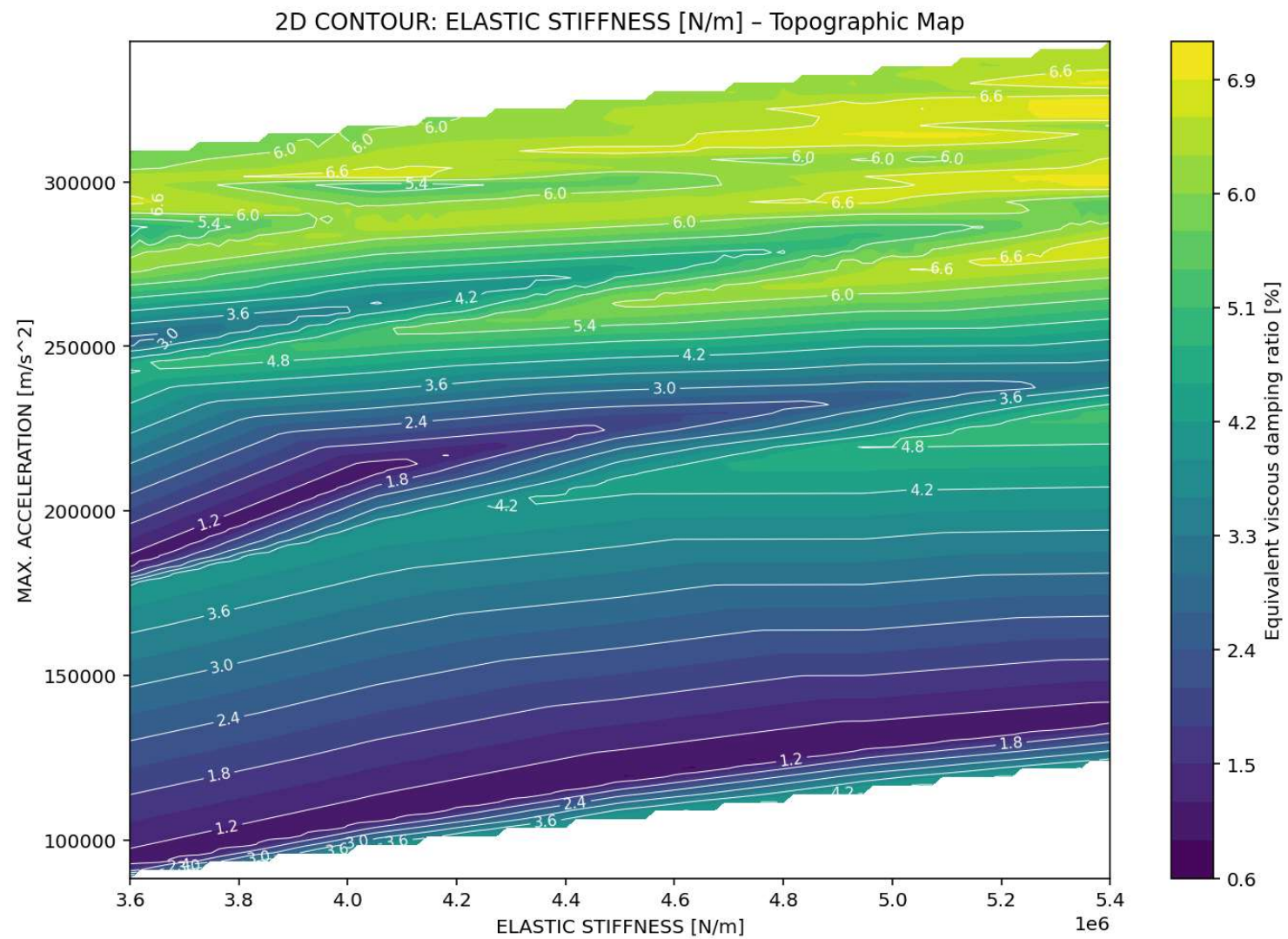
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. ACCELERATION [m/s²]



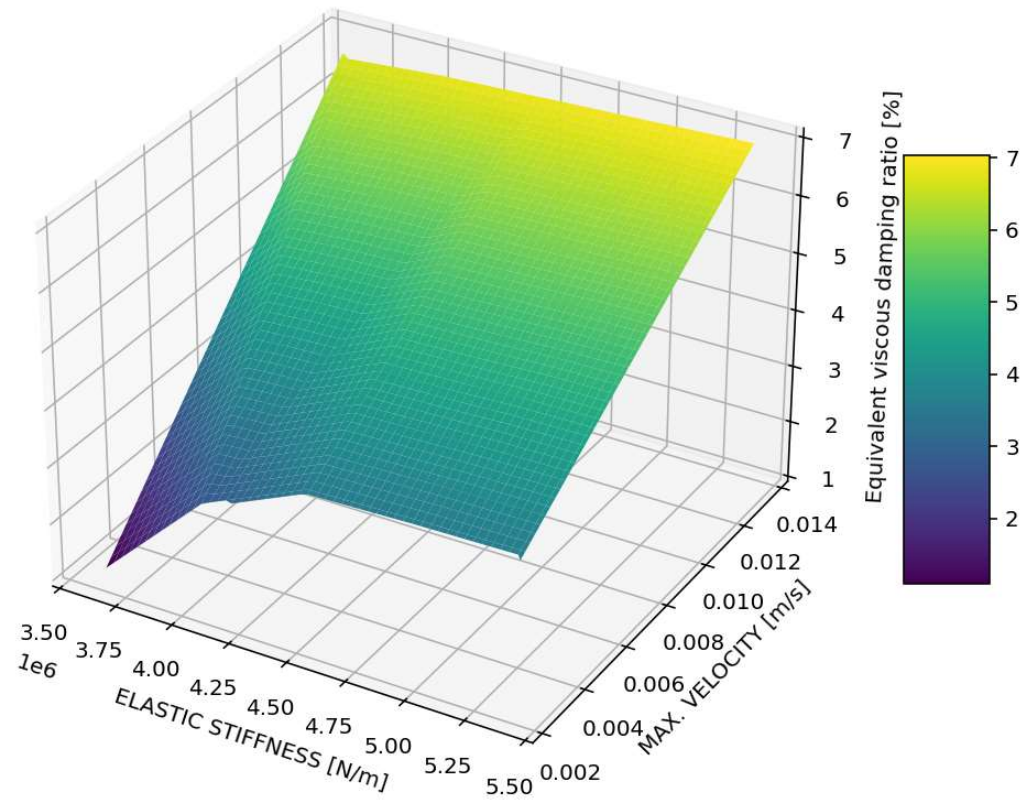


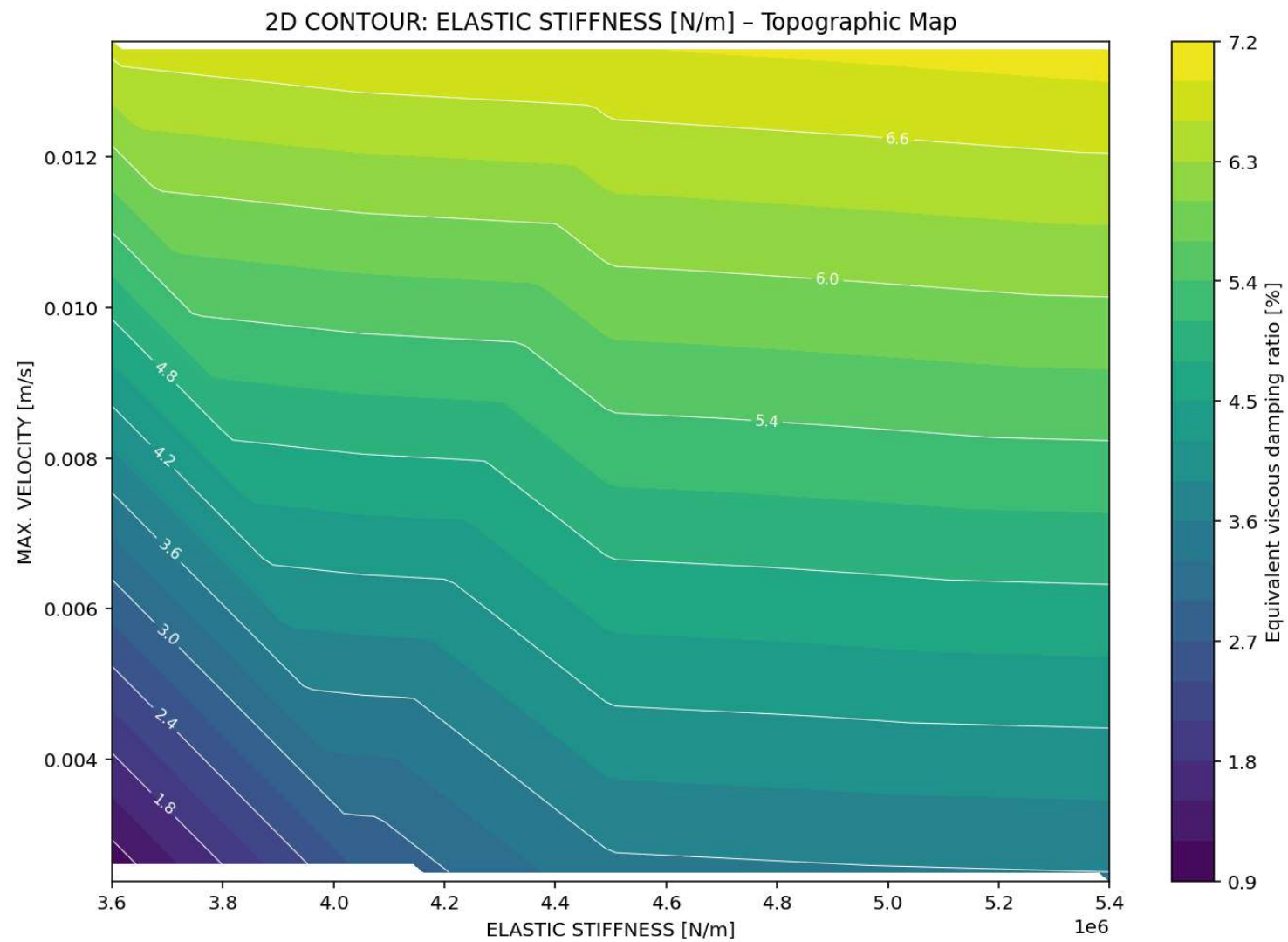
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. ACCELERATION [m/s²] - Equivalent viscous damping ratio [%]



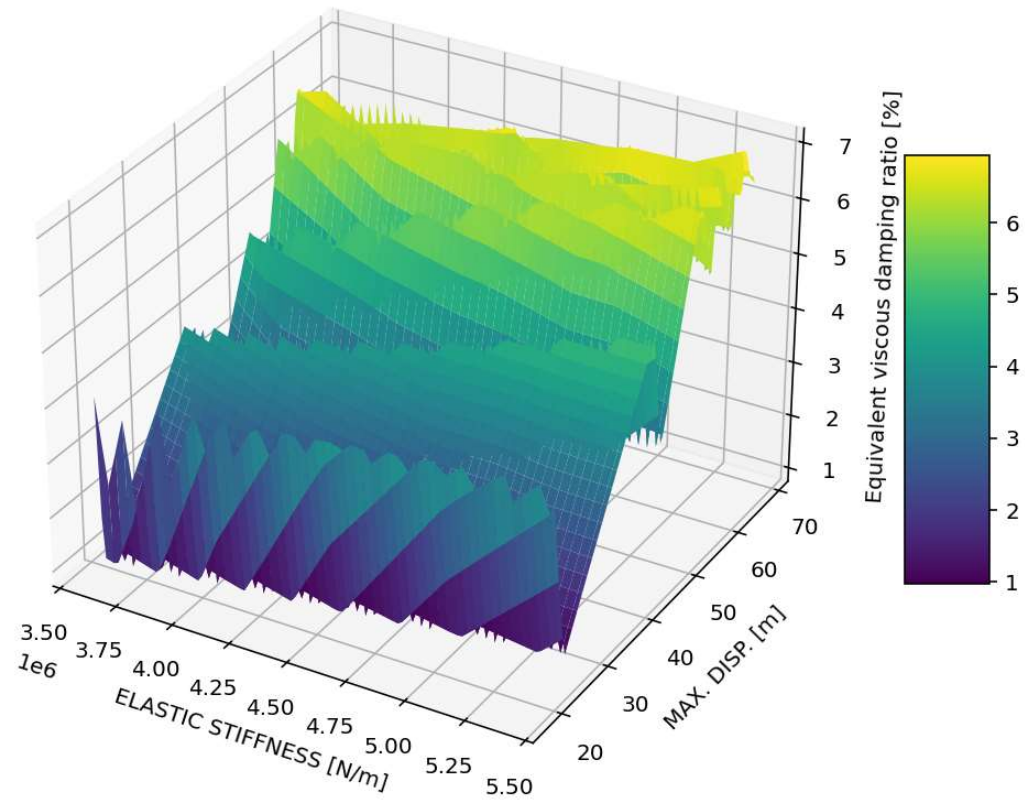


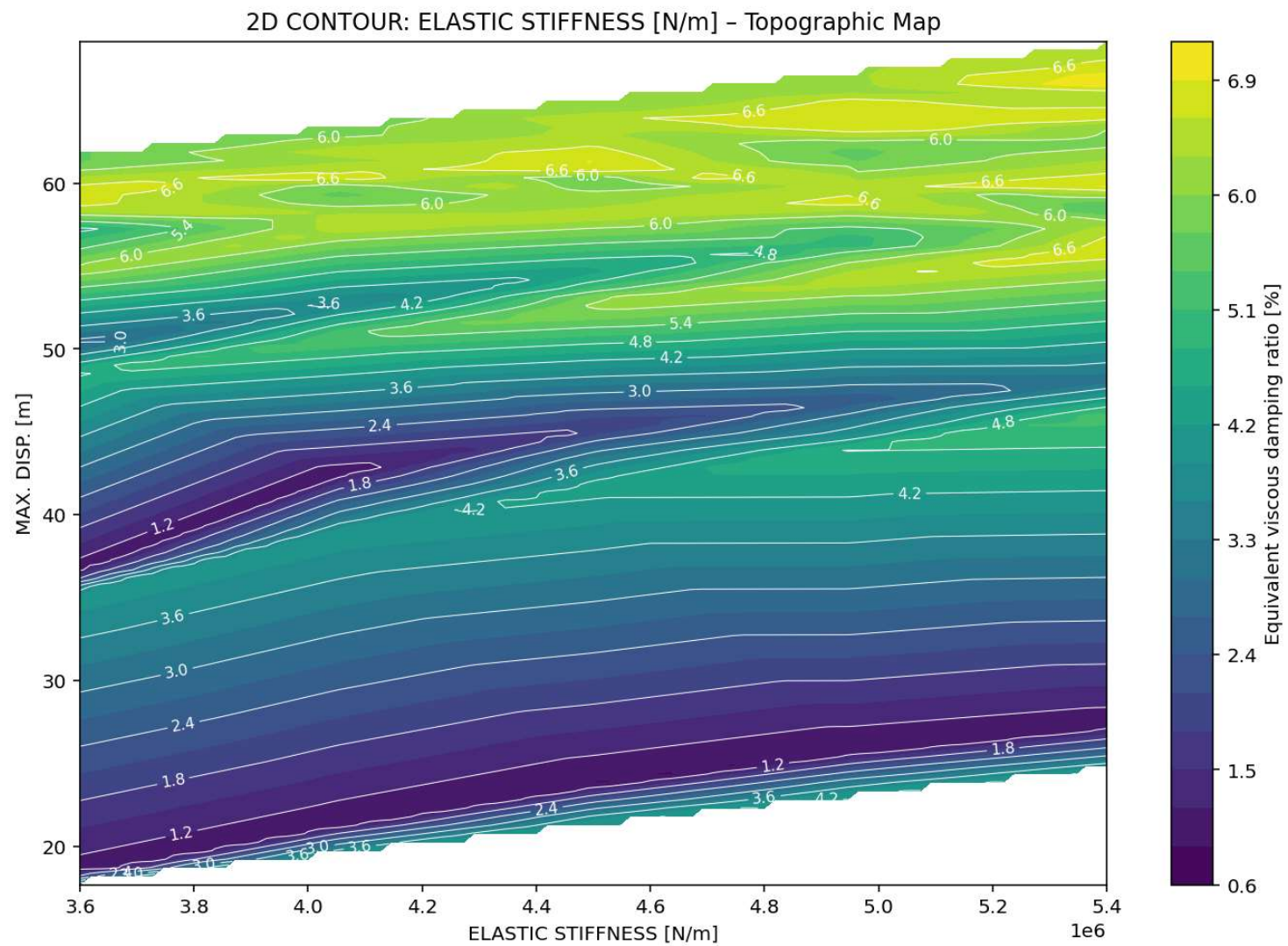
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. VELOCITY [m/s] - Equivalent viscous damping ratio [%]



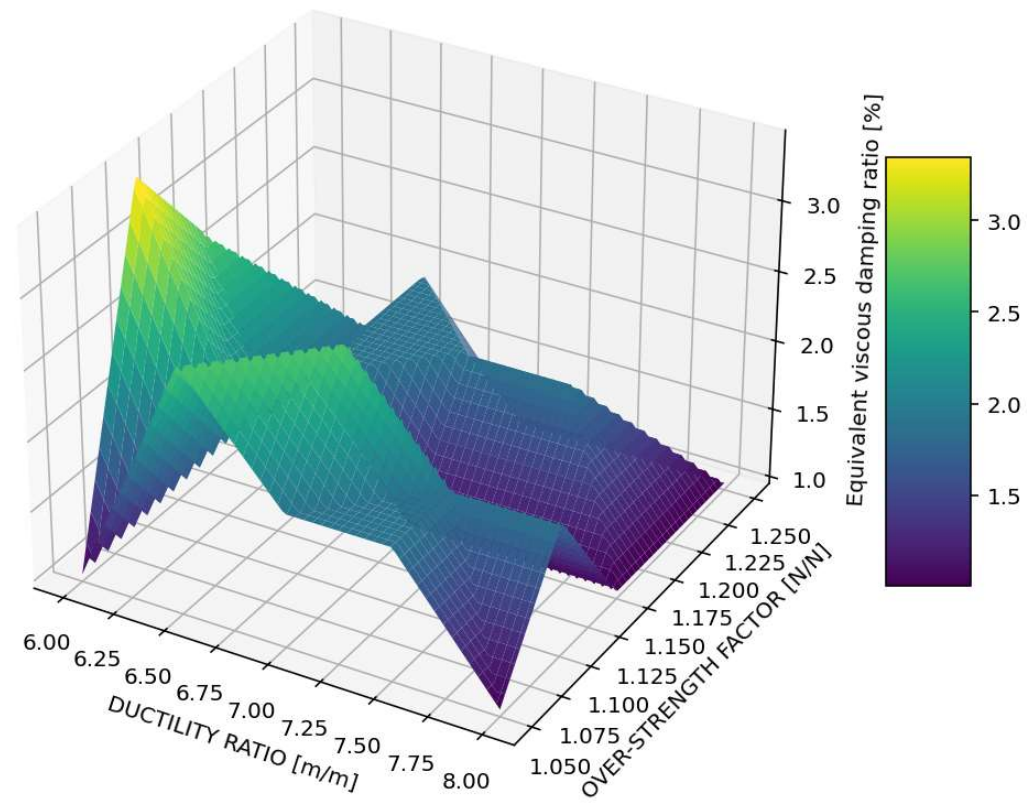


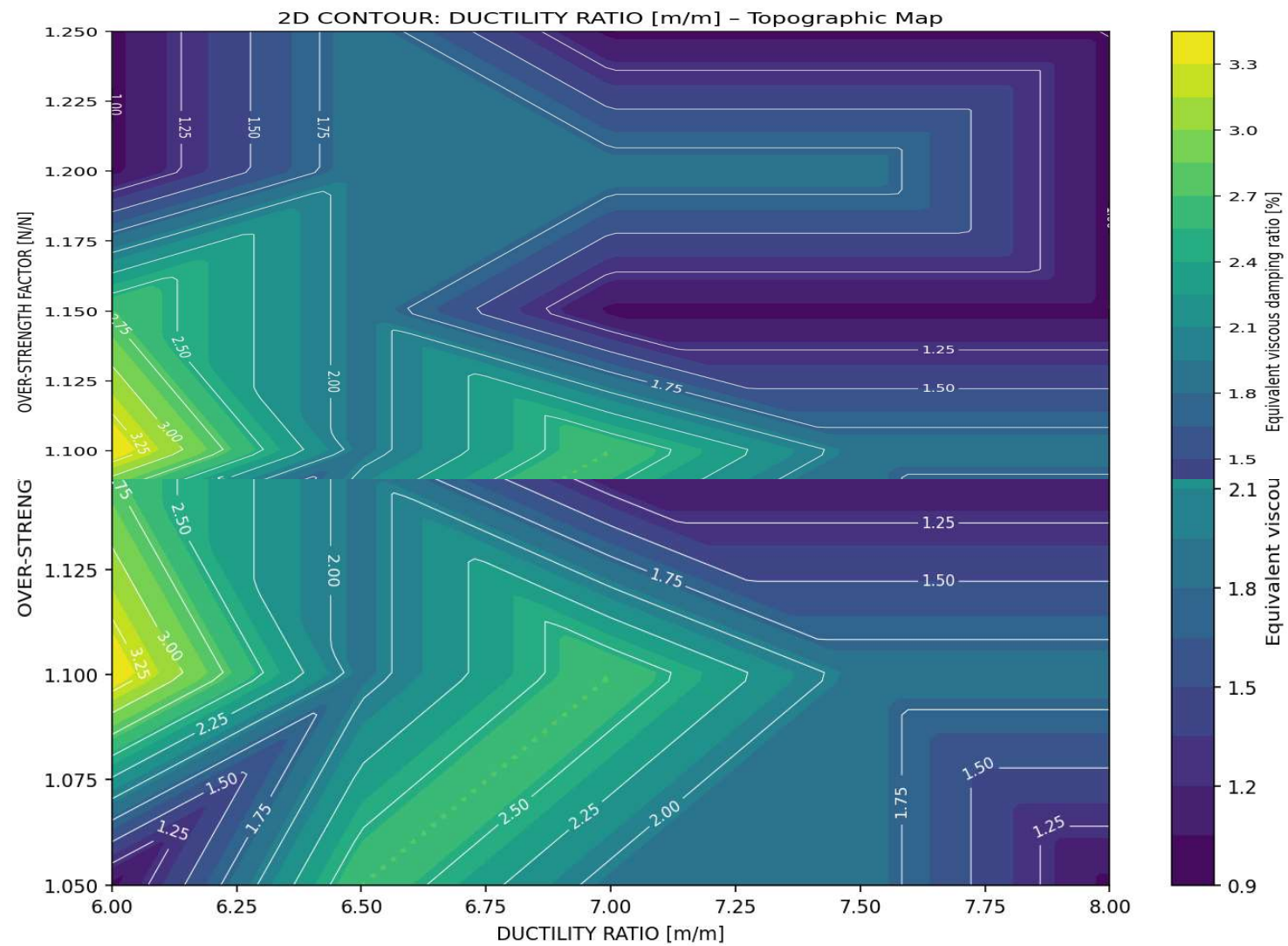
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. DISP. [m] - Equivalent viscous damping ratio [%]



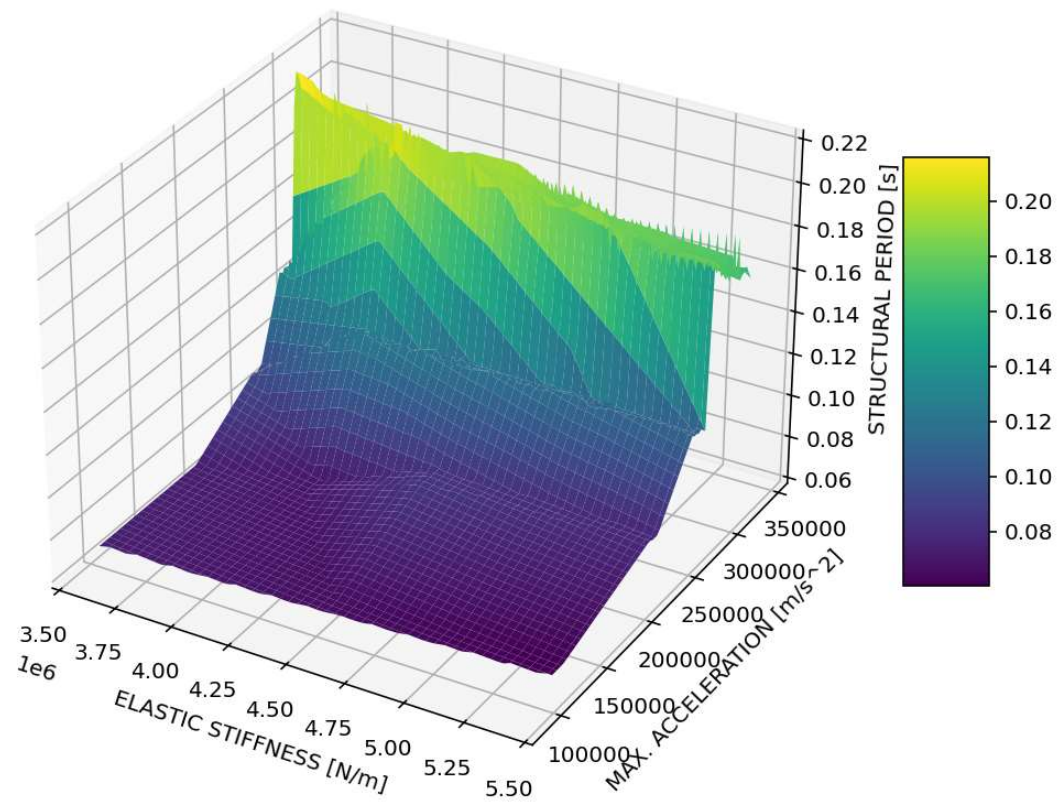


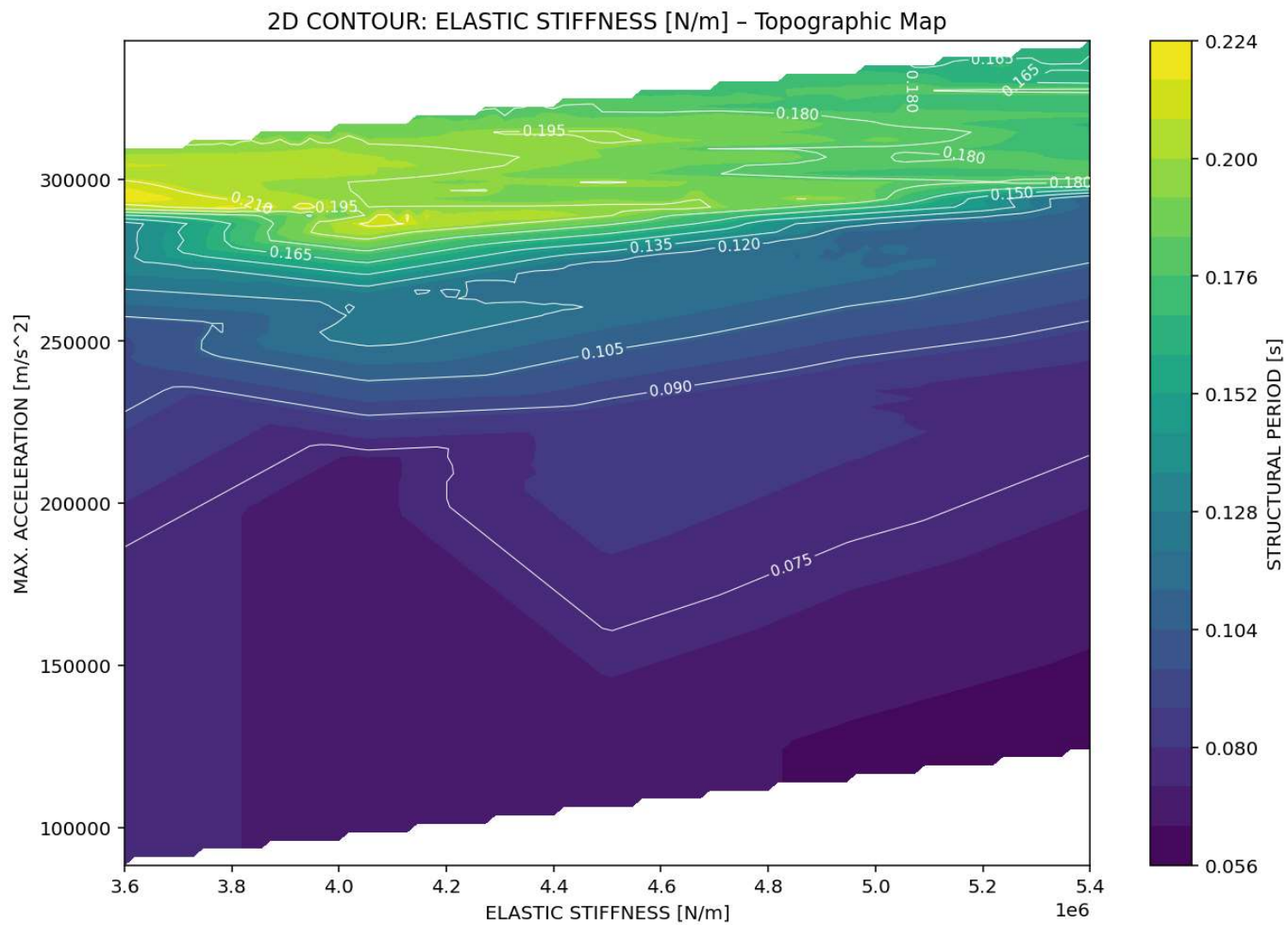
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - Equivalent viscous damping ratio [%]



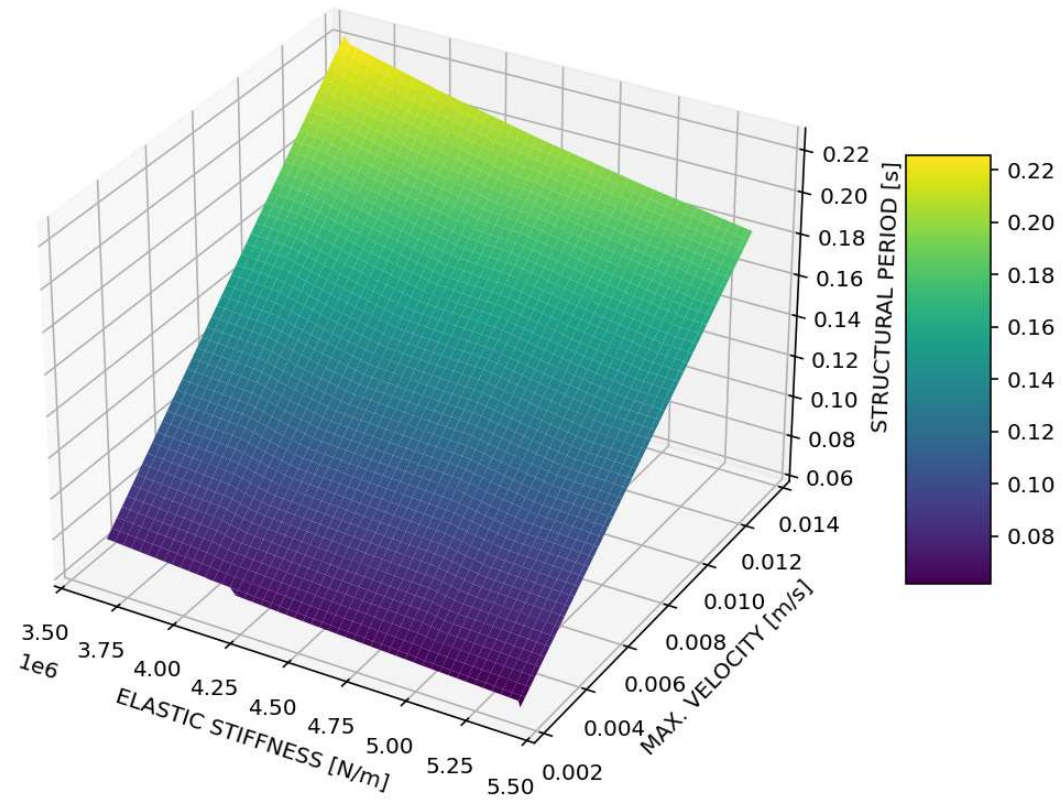


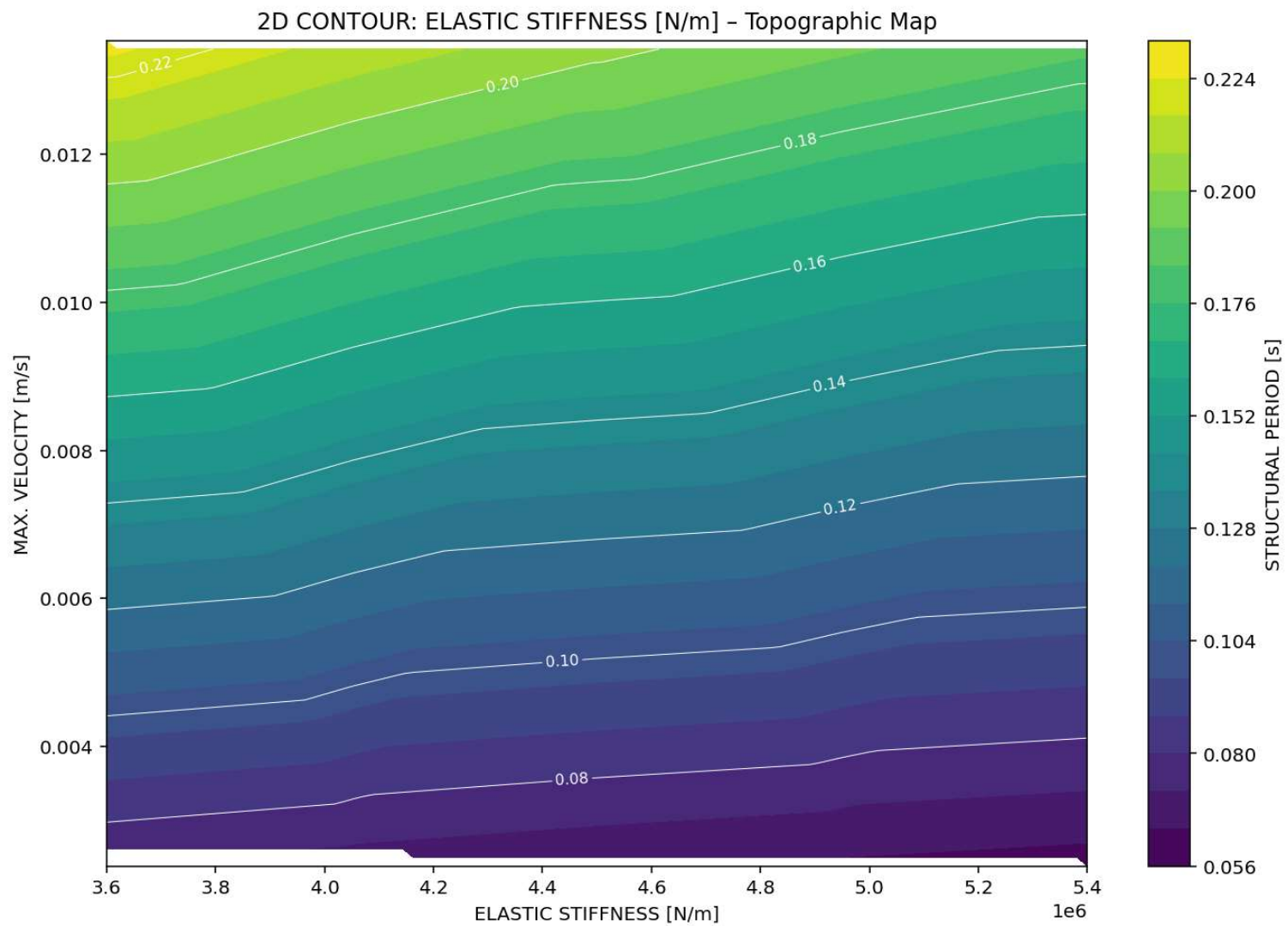
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. ACCELERATION [m/s²] - STRUCTURAL PERIOD [s]



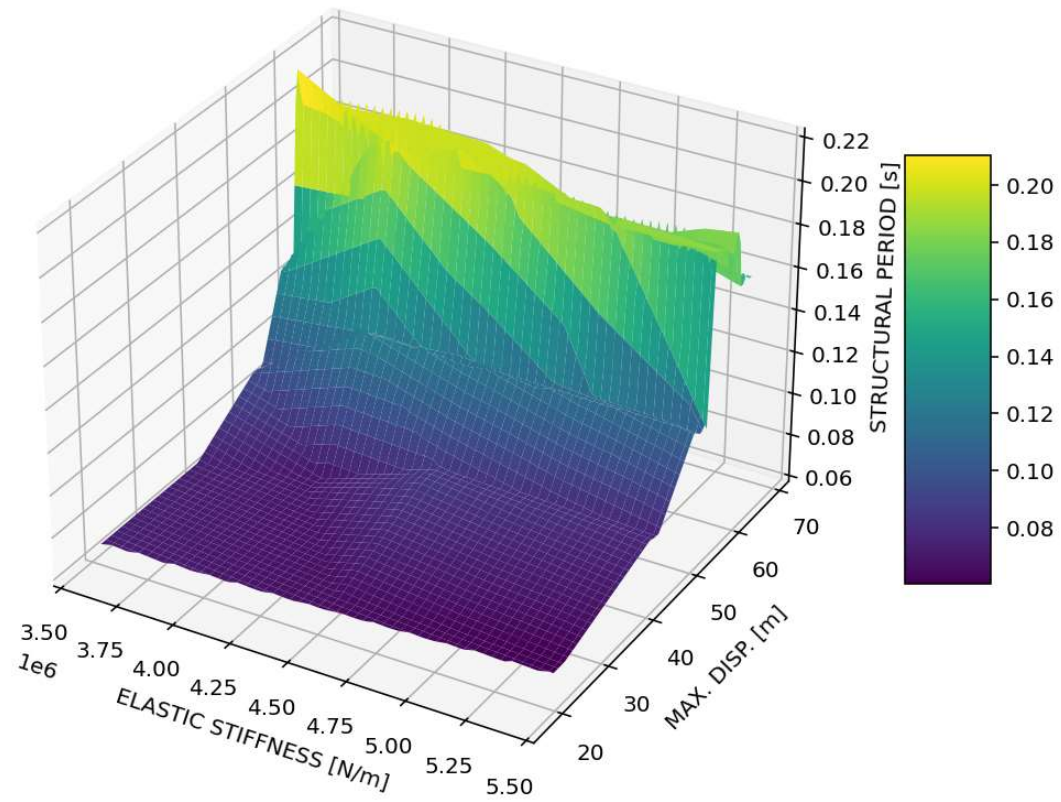


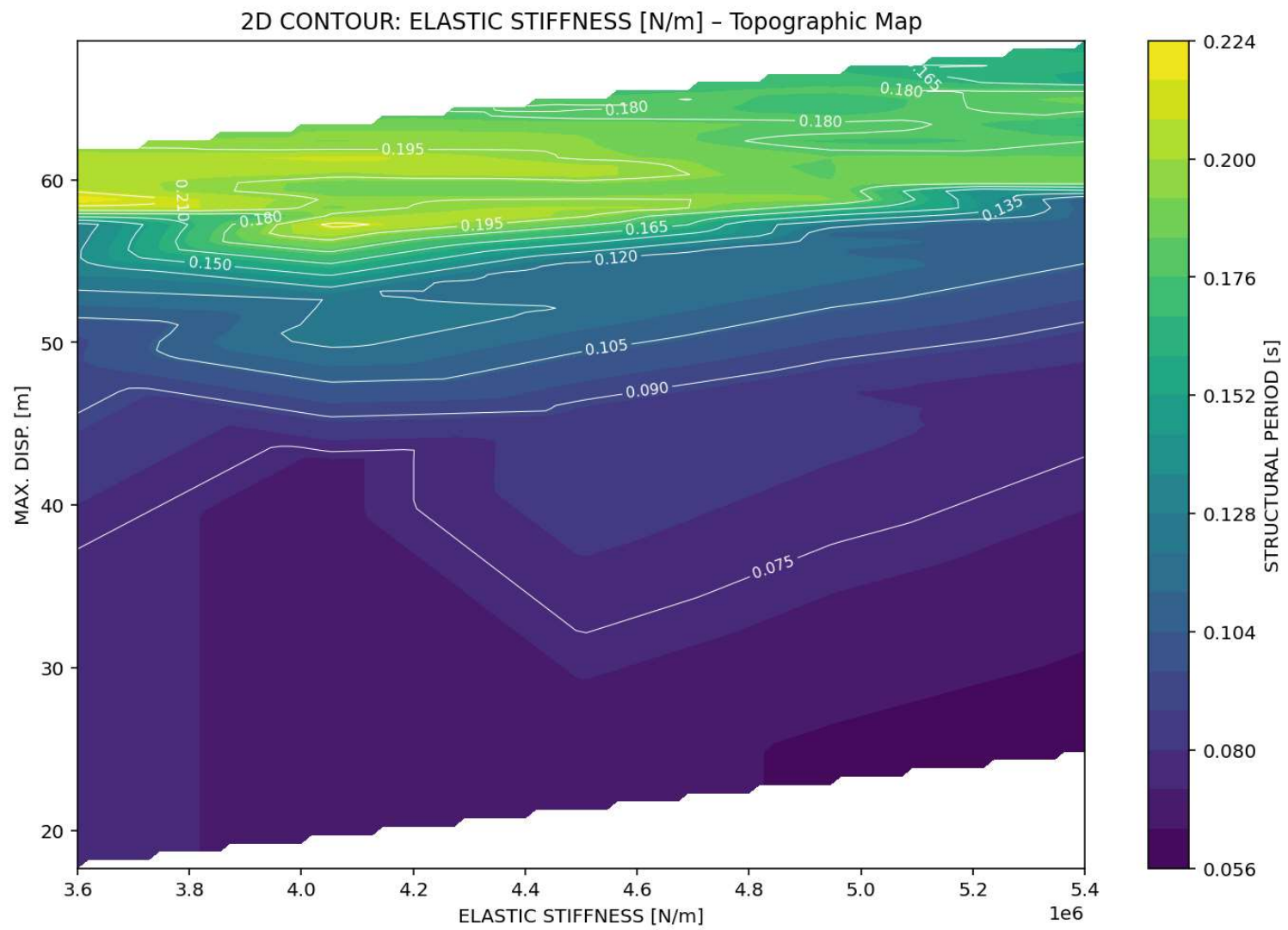
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. VELOCITY [m/s] - STRUCTURAL PERIOD [s]



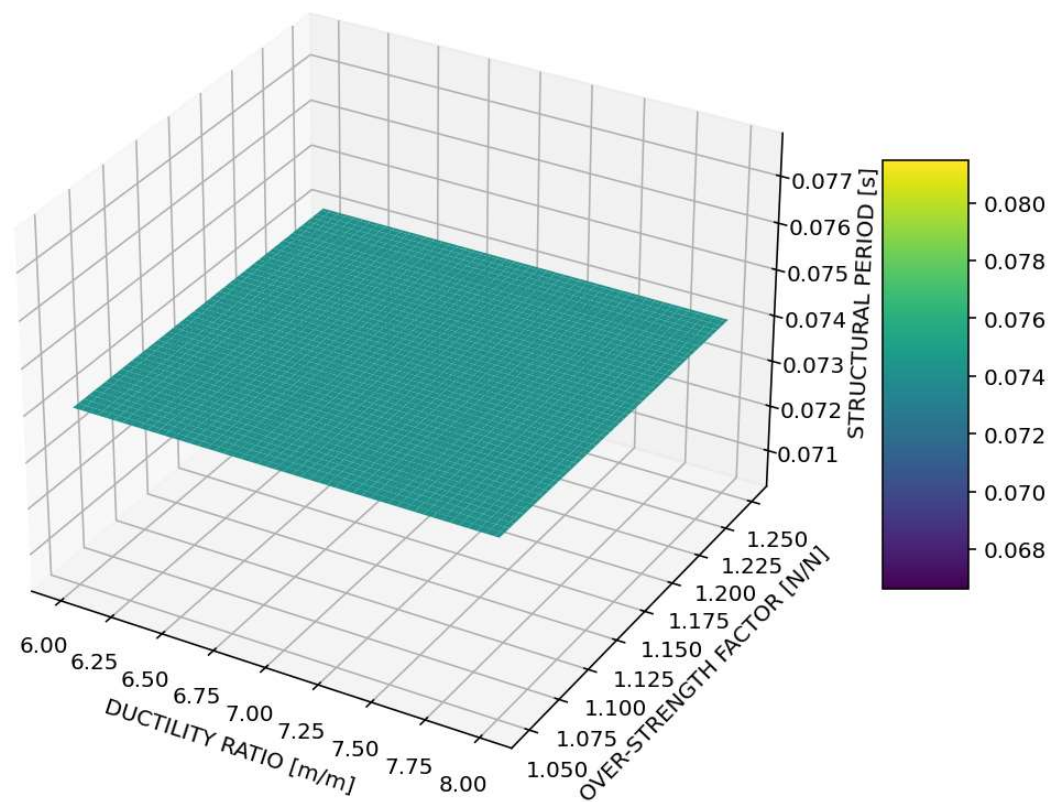


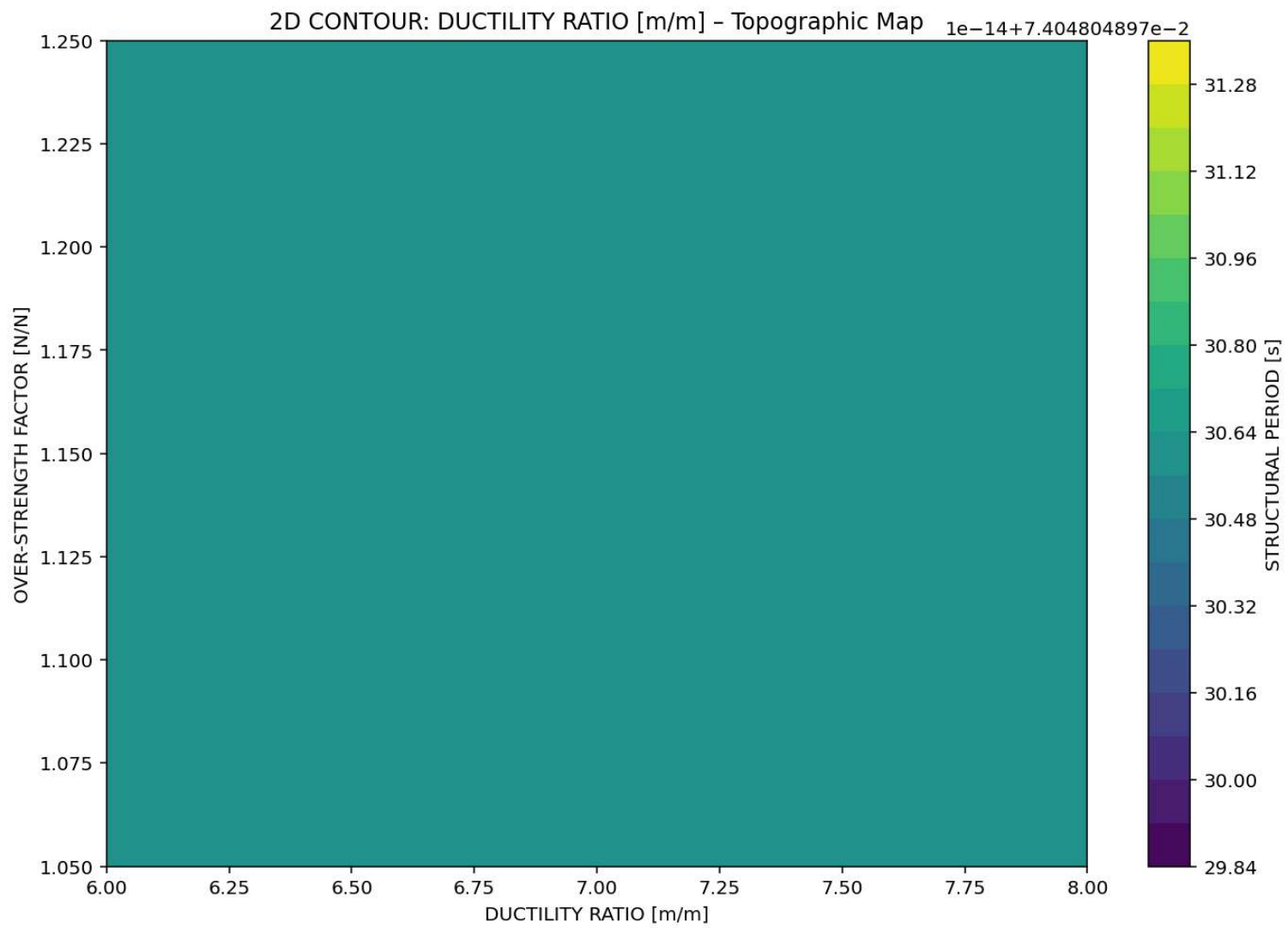
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. DISP. [m] - STRUCTURAL PERIOD [s]



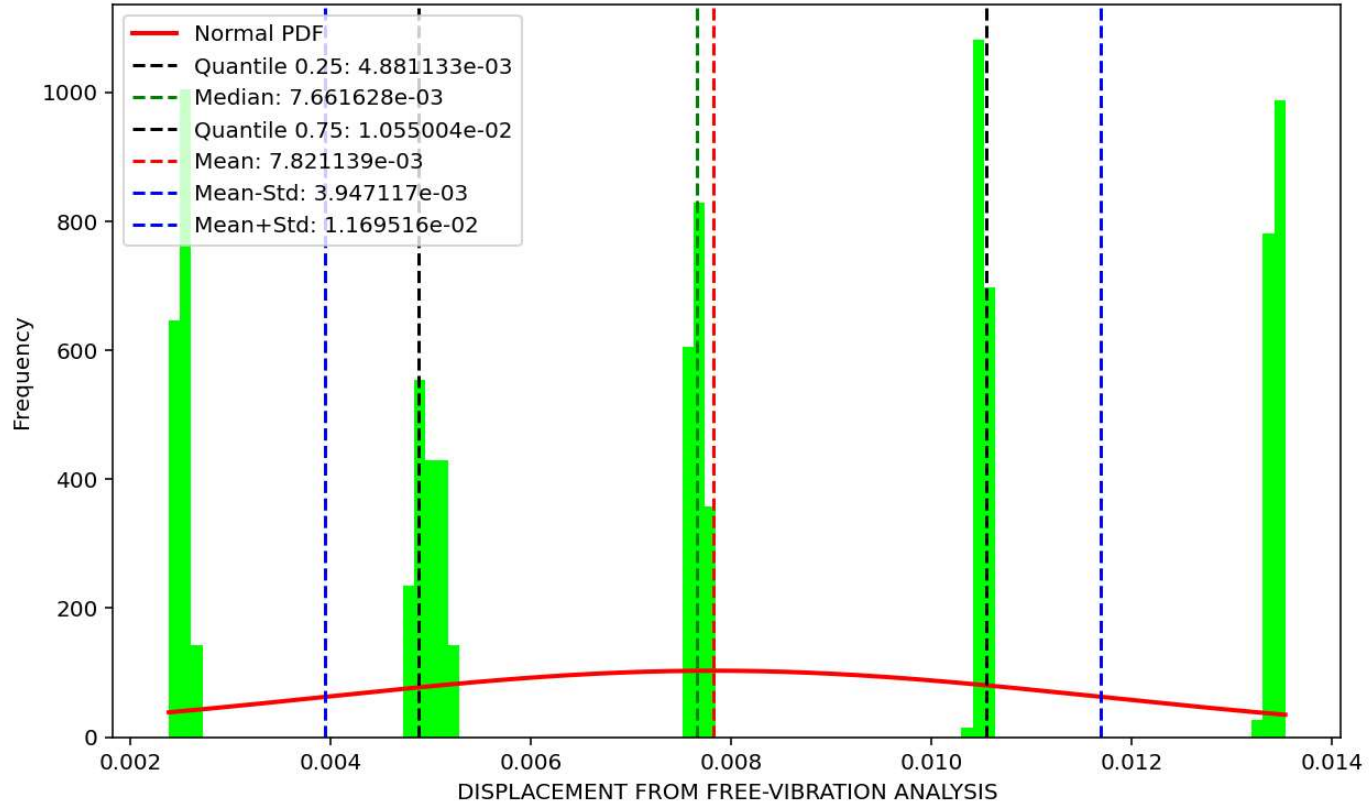


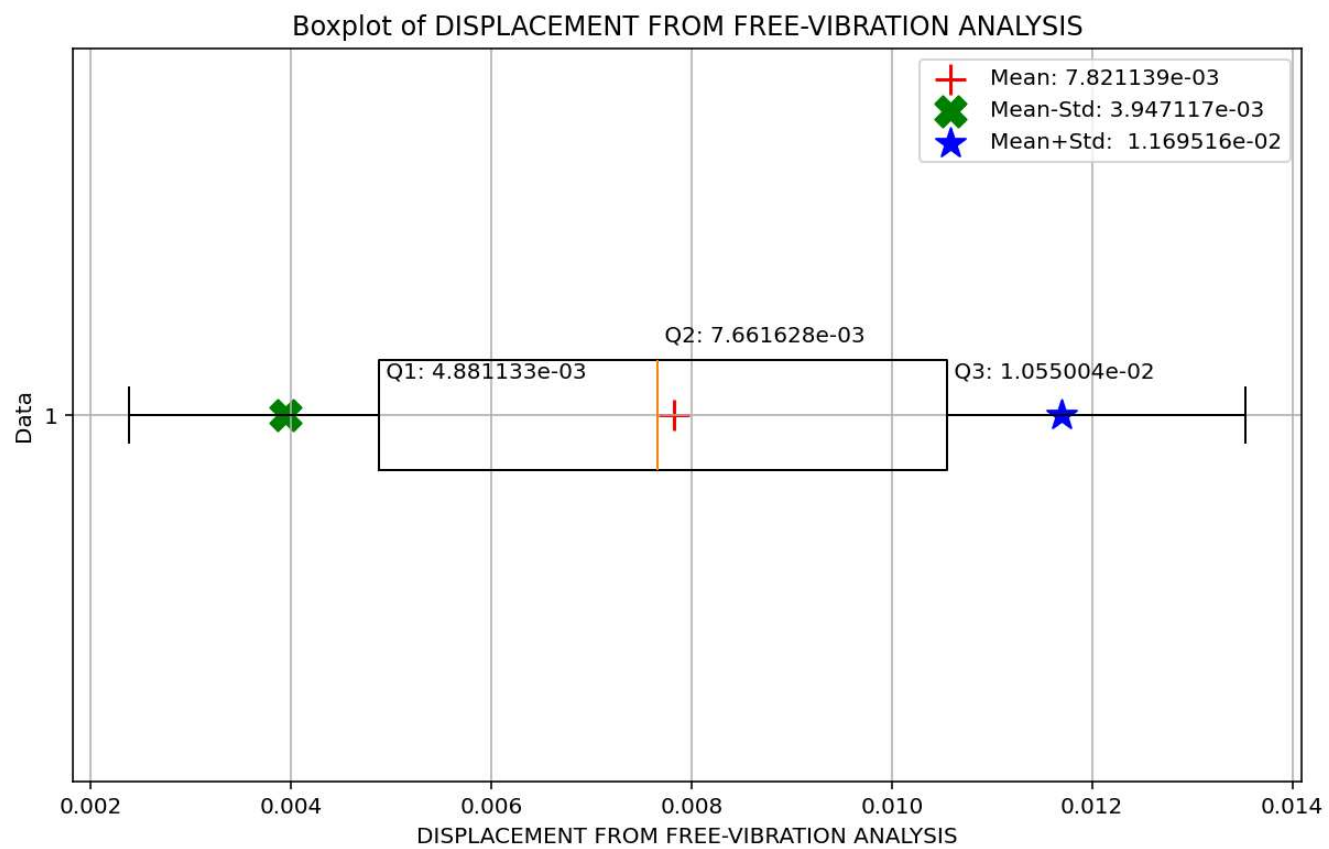
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - STRUCTURAL PERIOD [s]

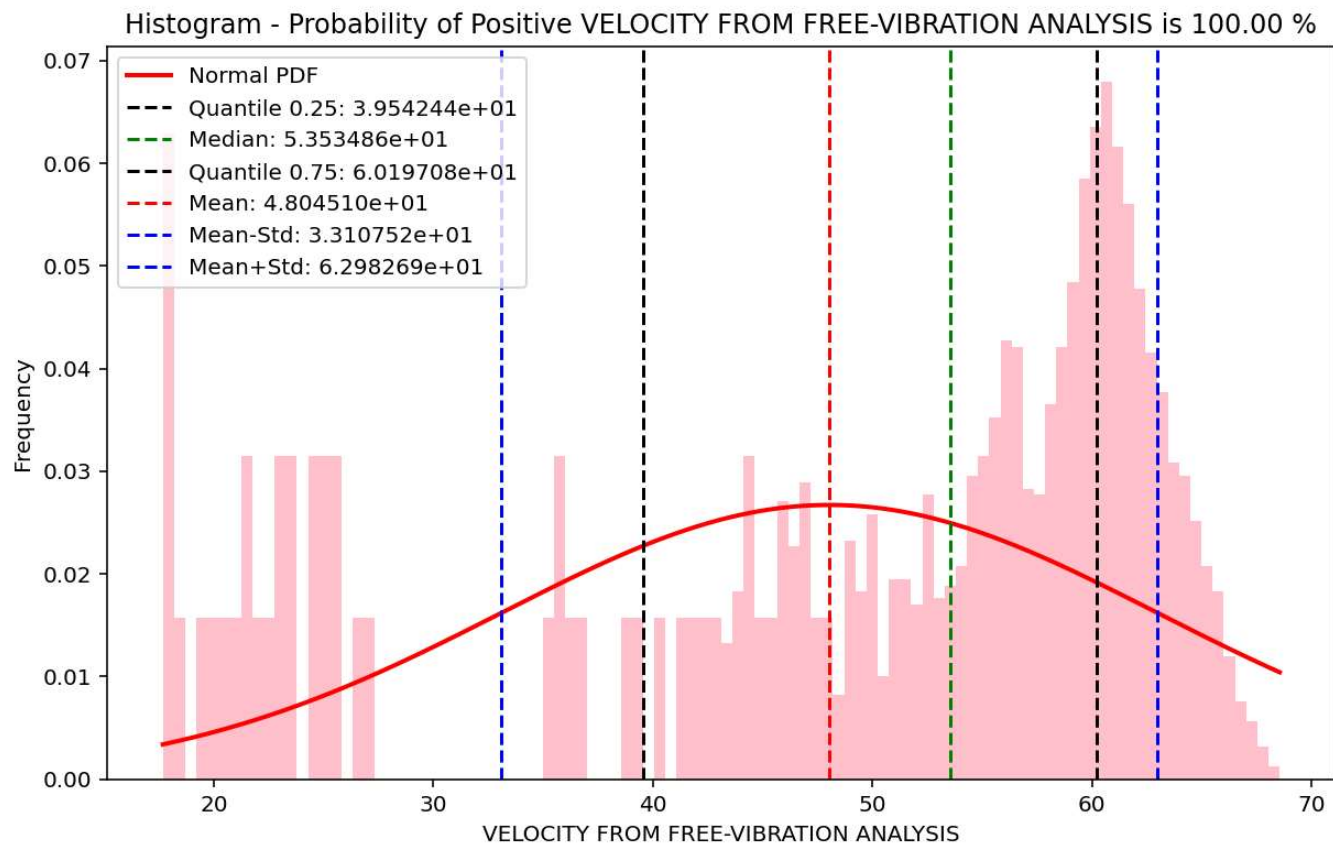




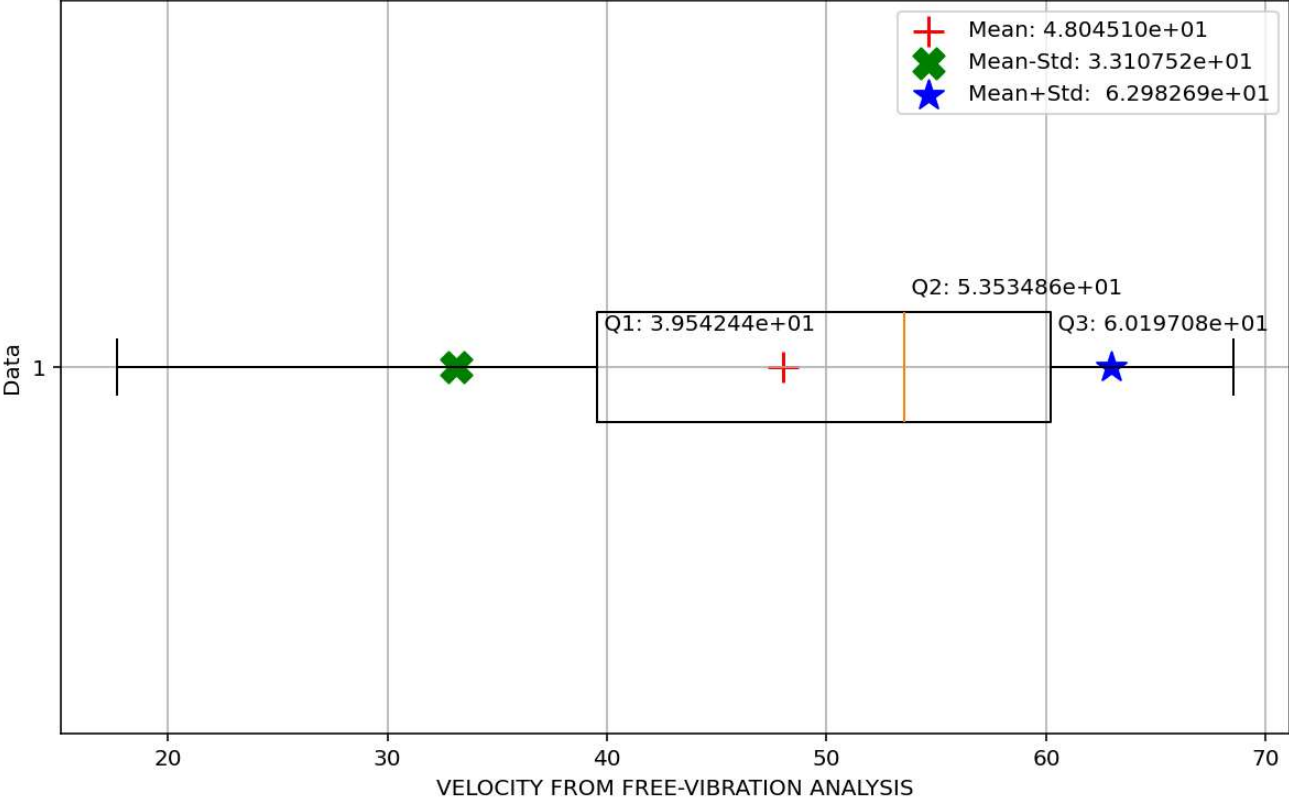
Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 100.00 %



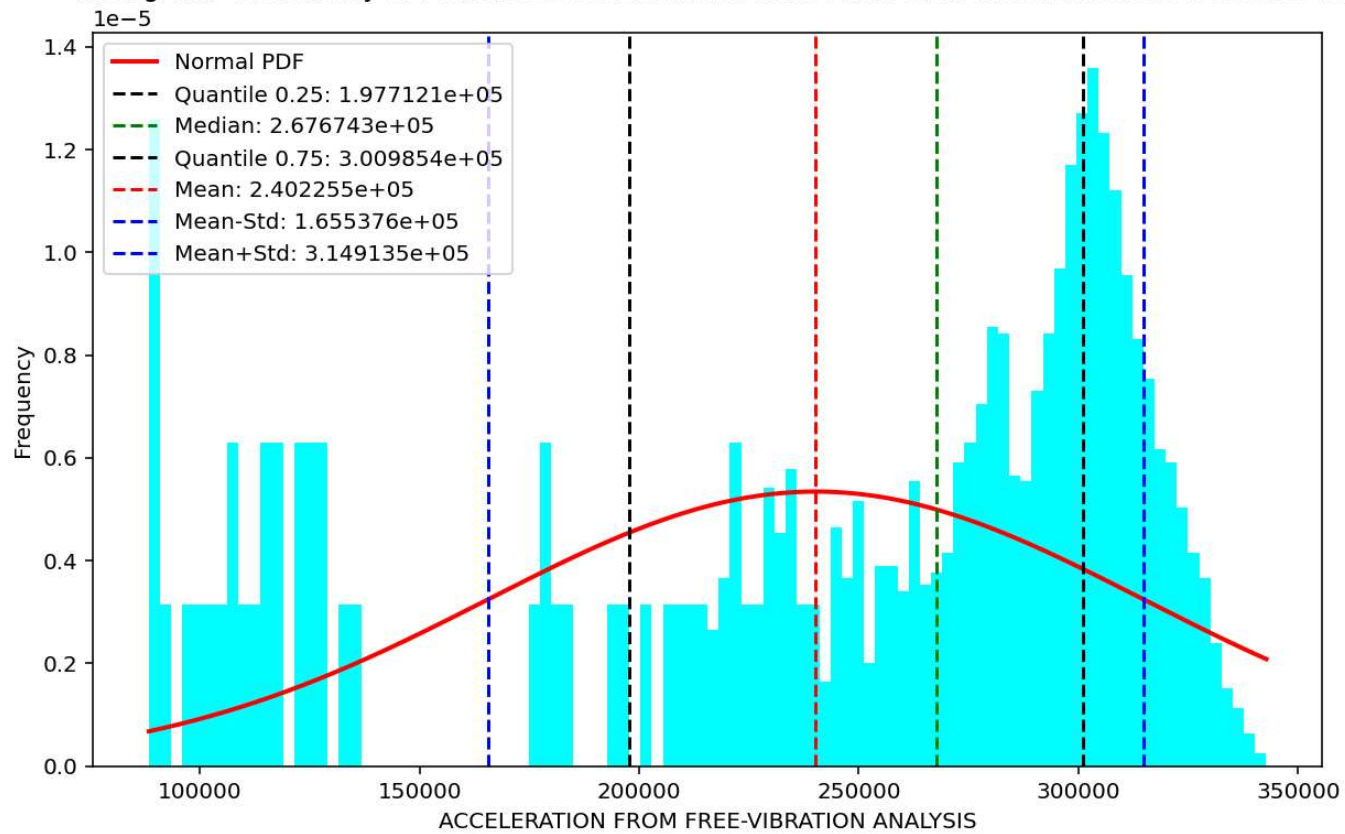


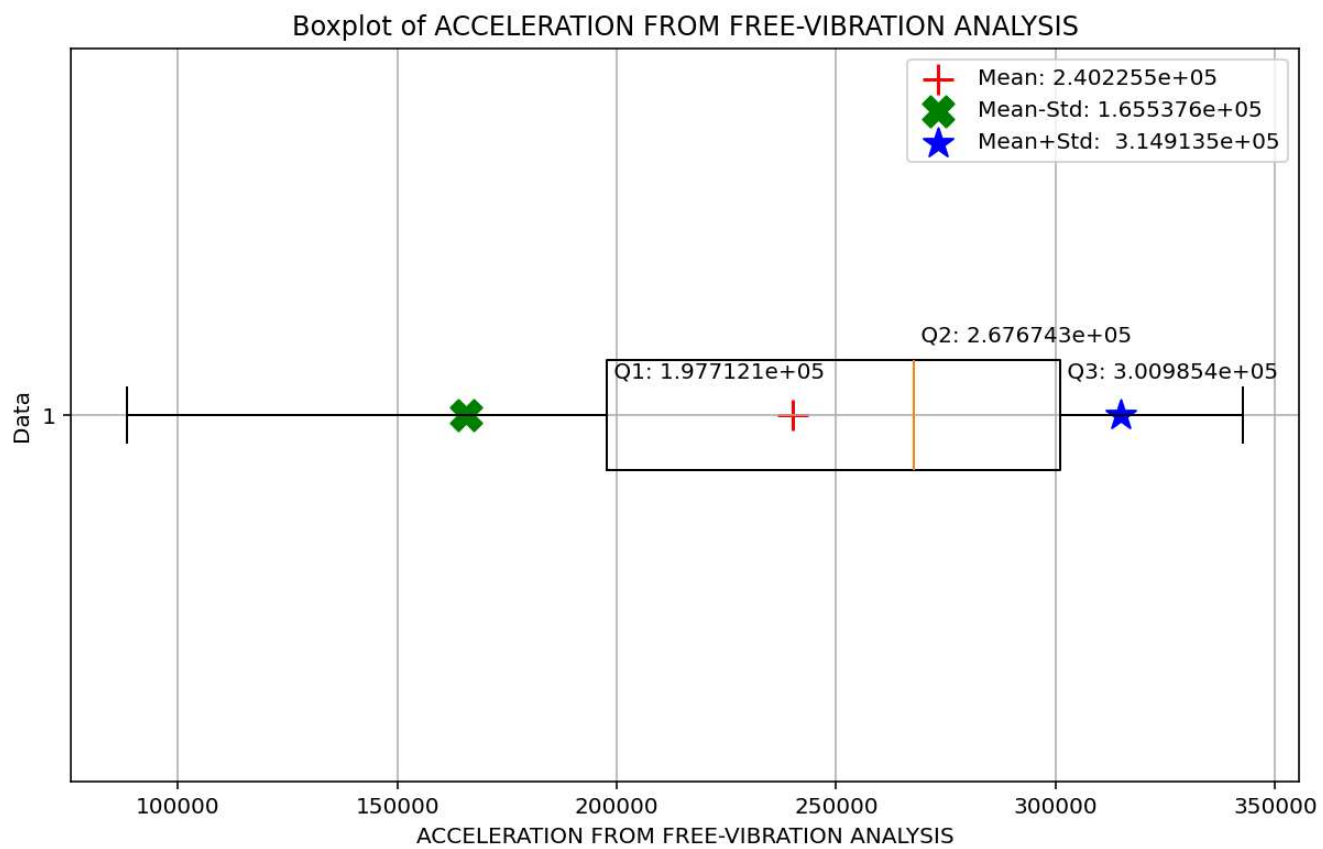


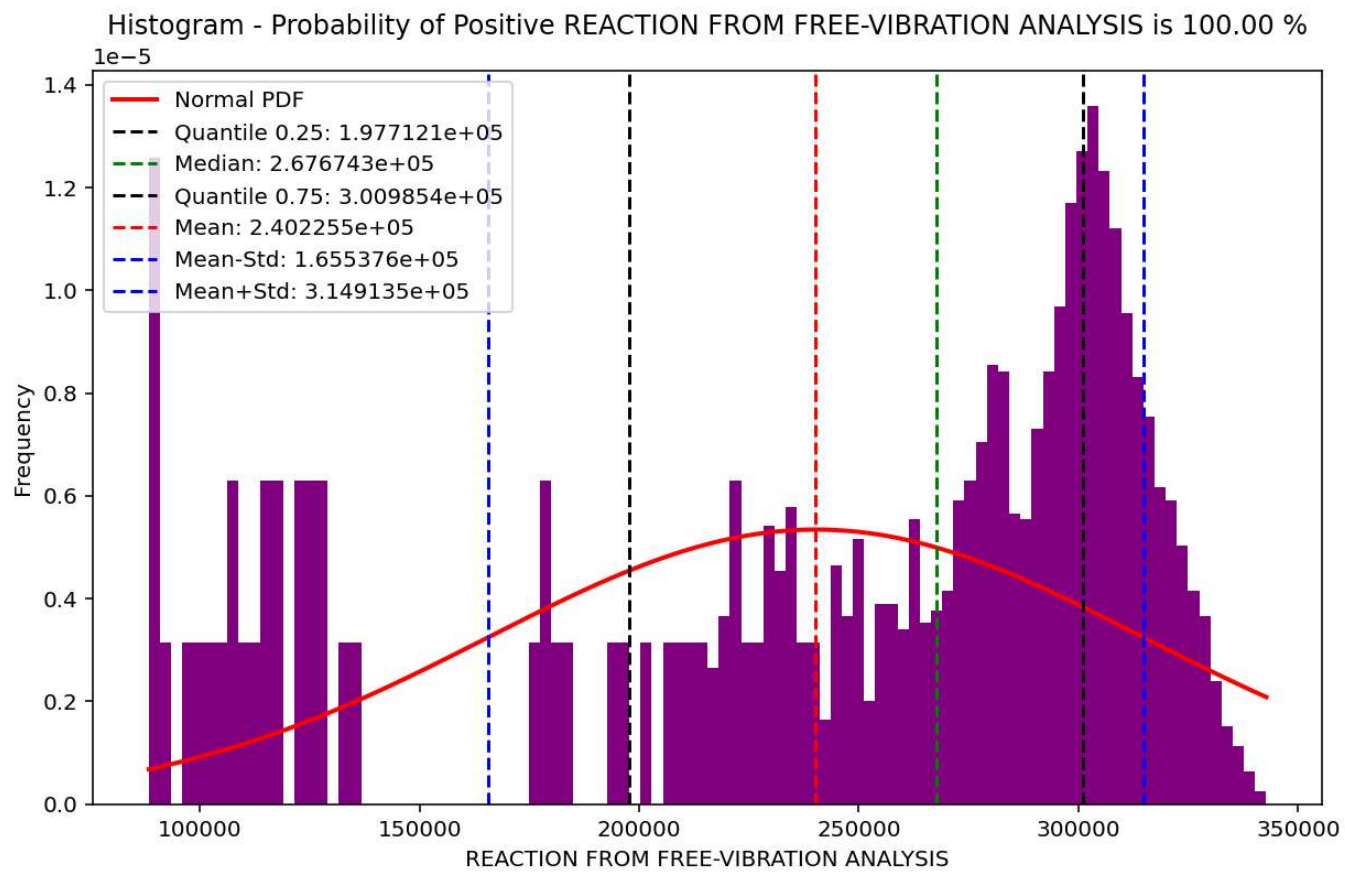
Boxplot of VELOCITY FROM FREE-VIBRATION ANALYSIS

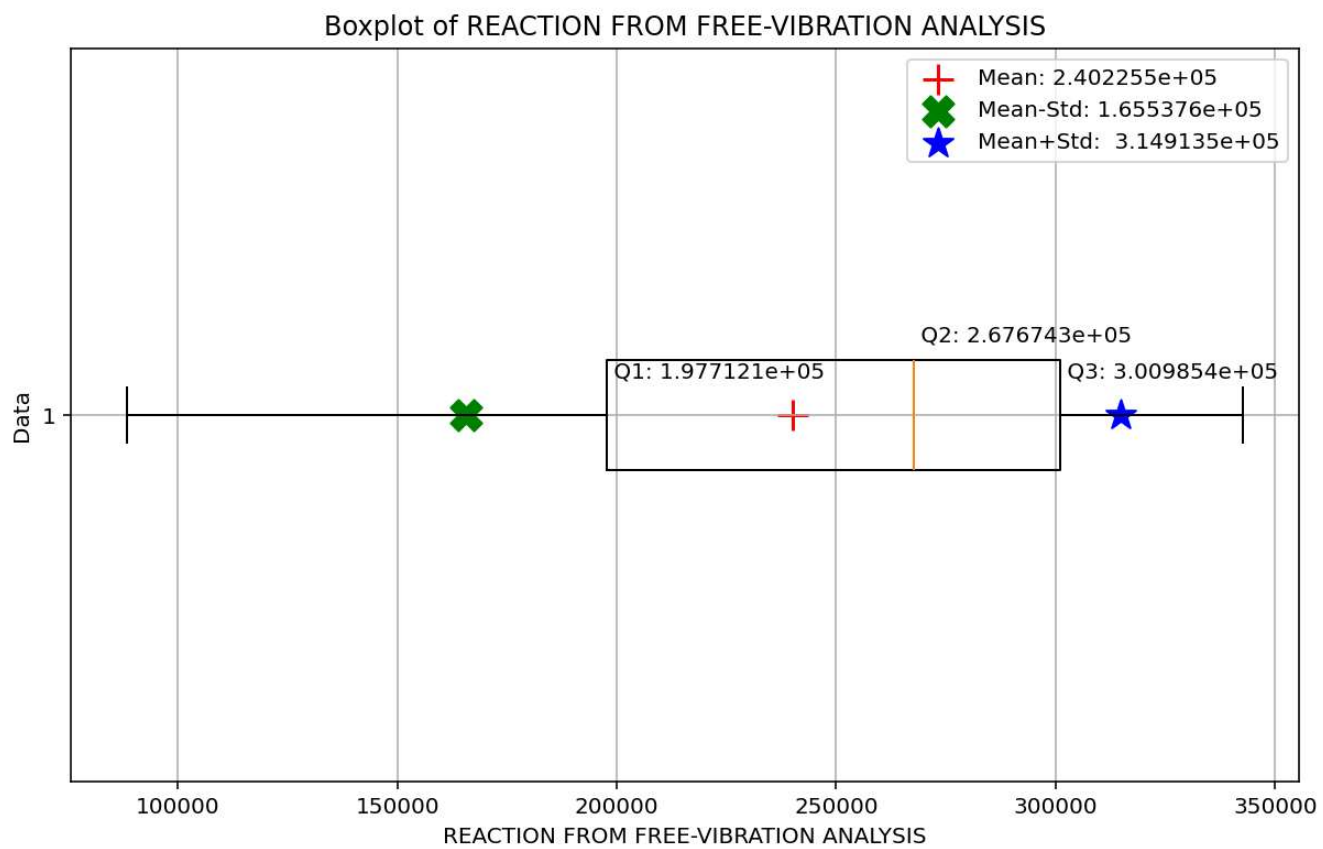


Histogram - Probability of Positive ACCELERATION FROM FREE-VIBRATION ANALYSIS is 100.00 %

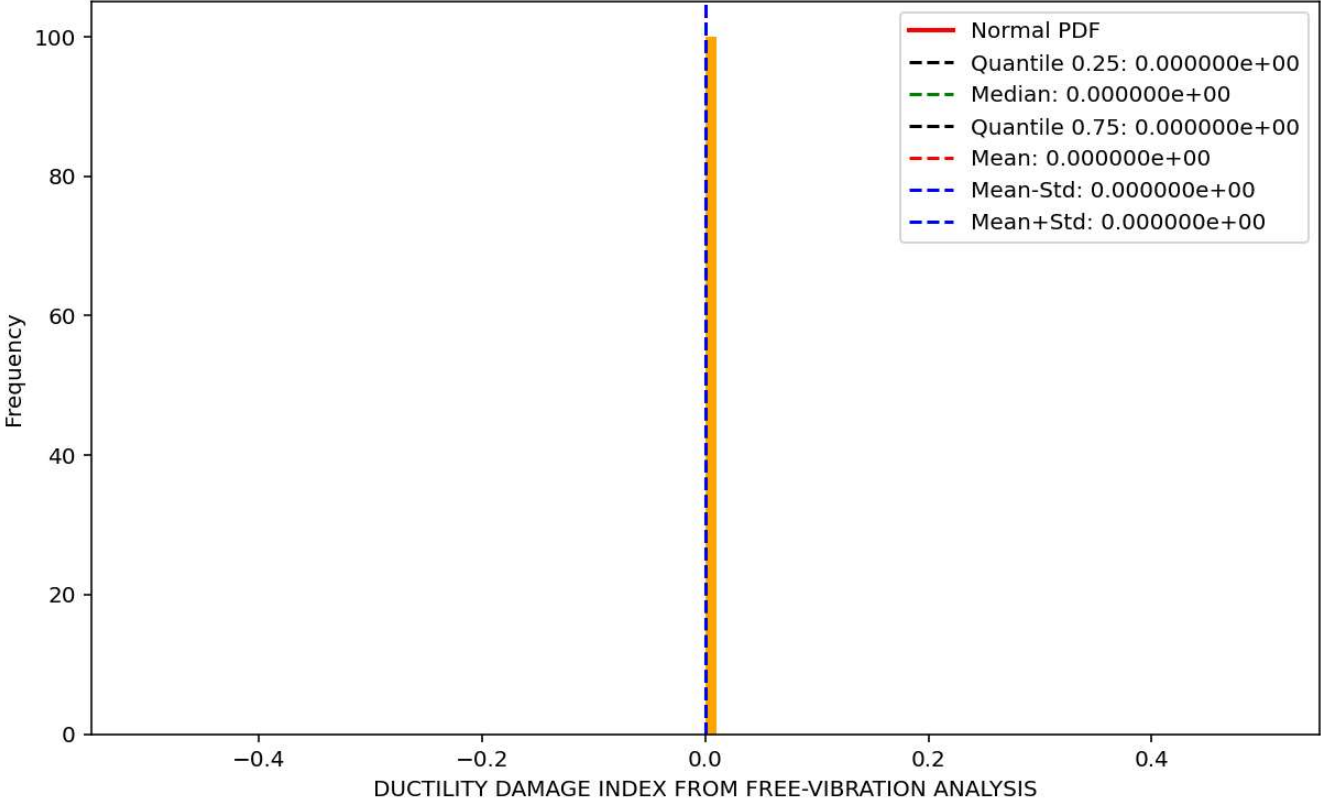


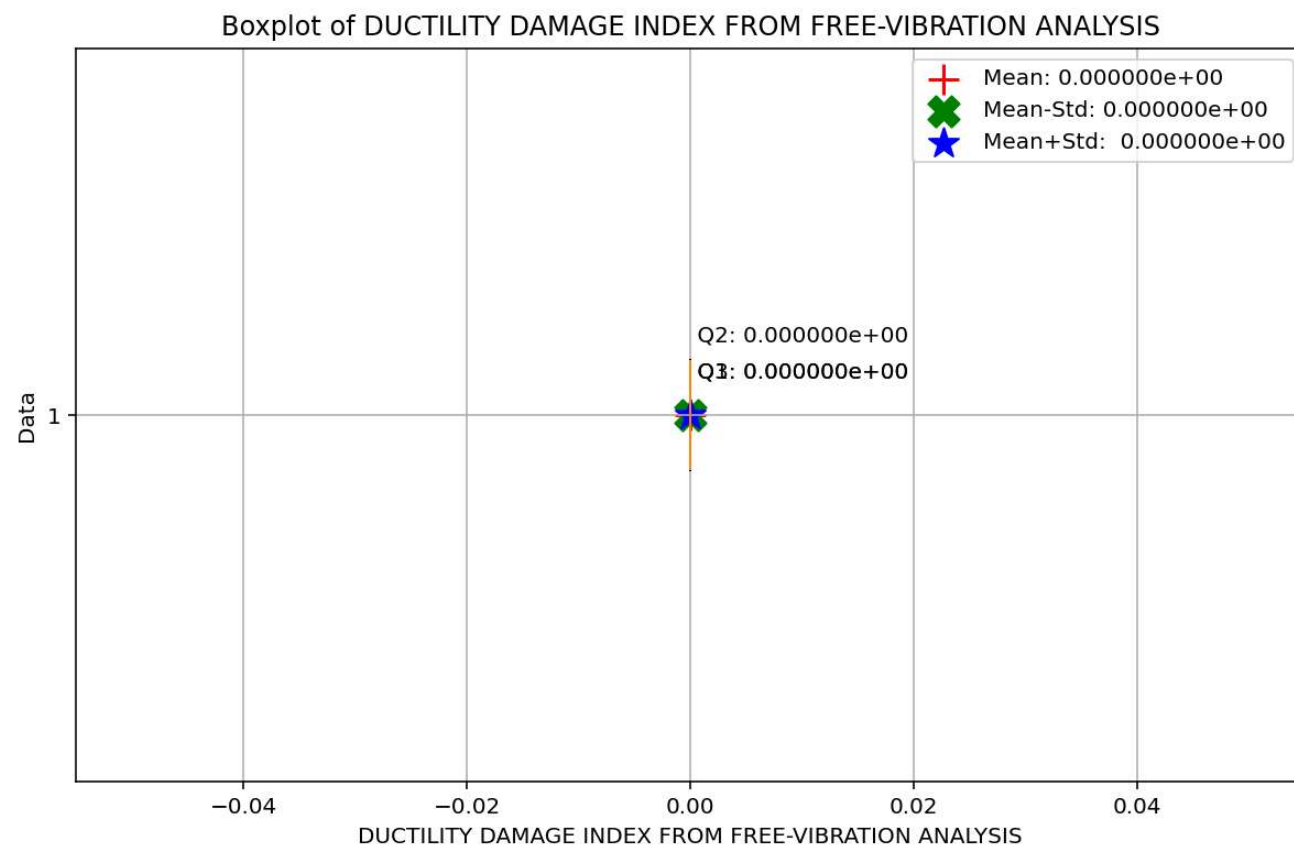




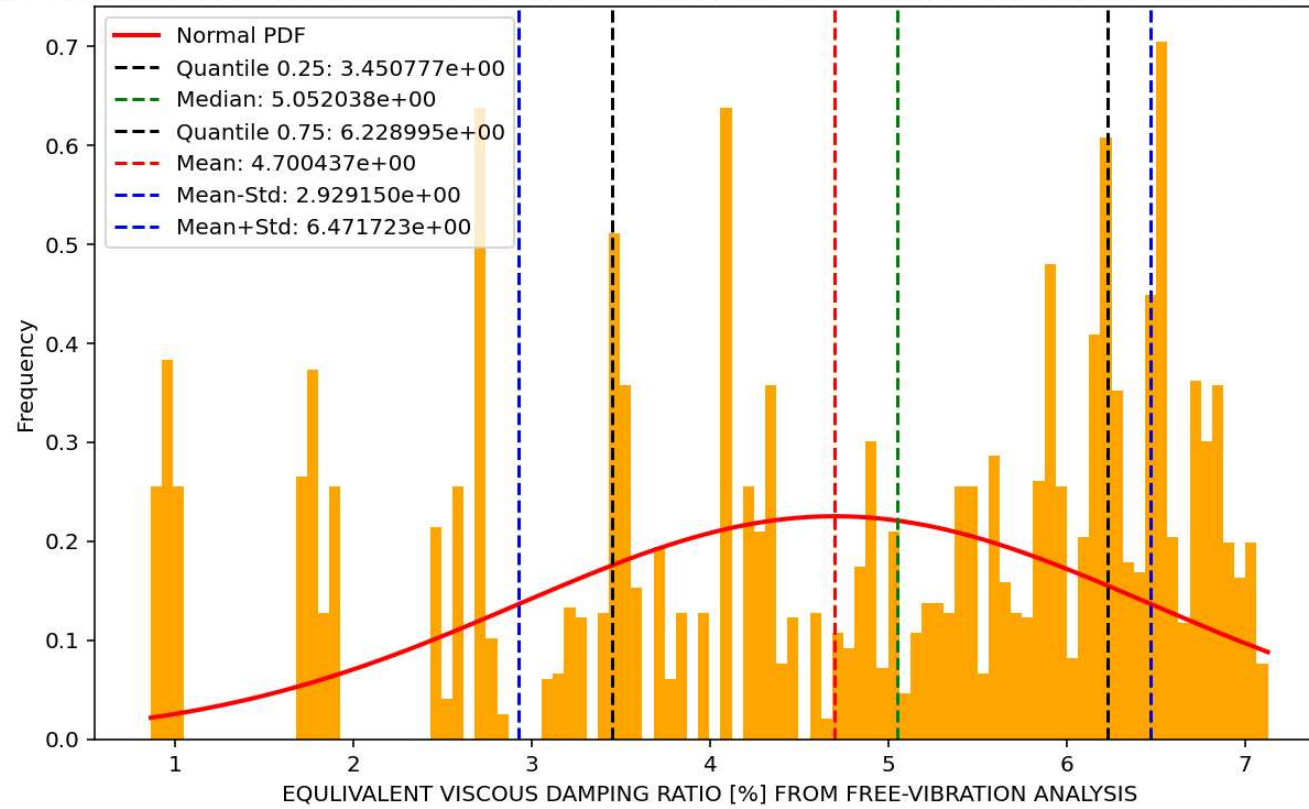


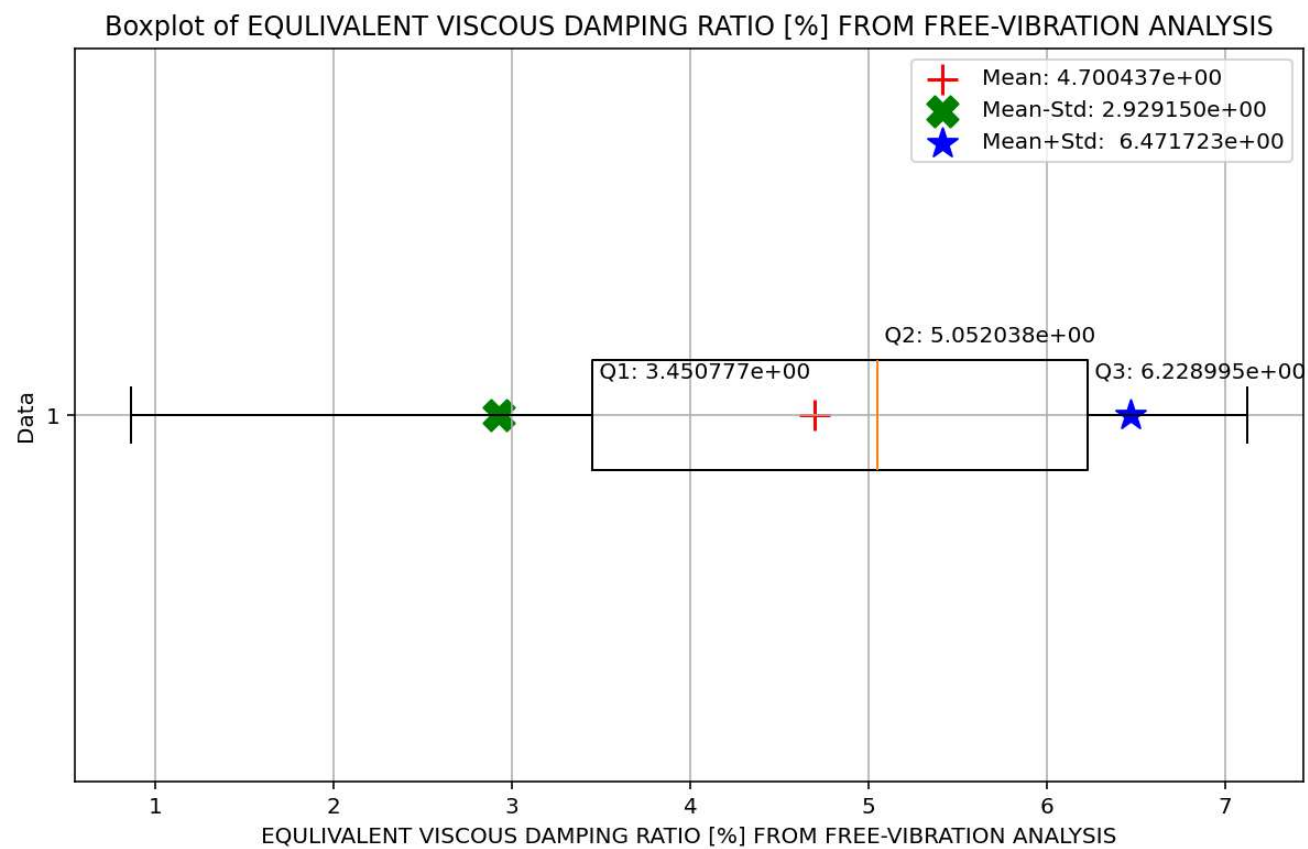
Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %



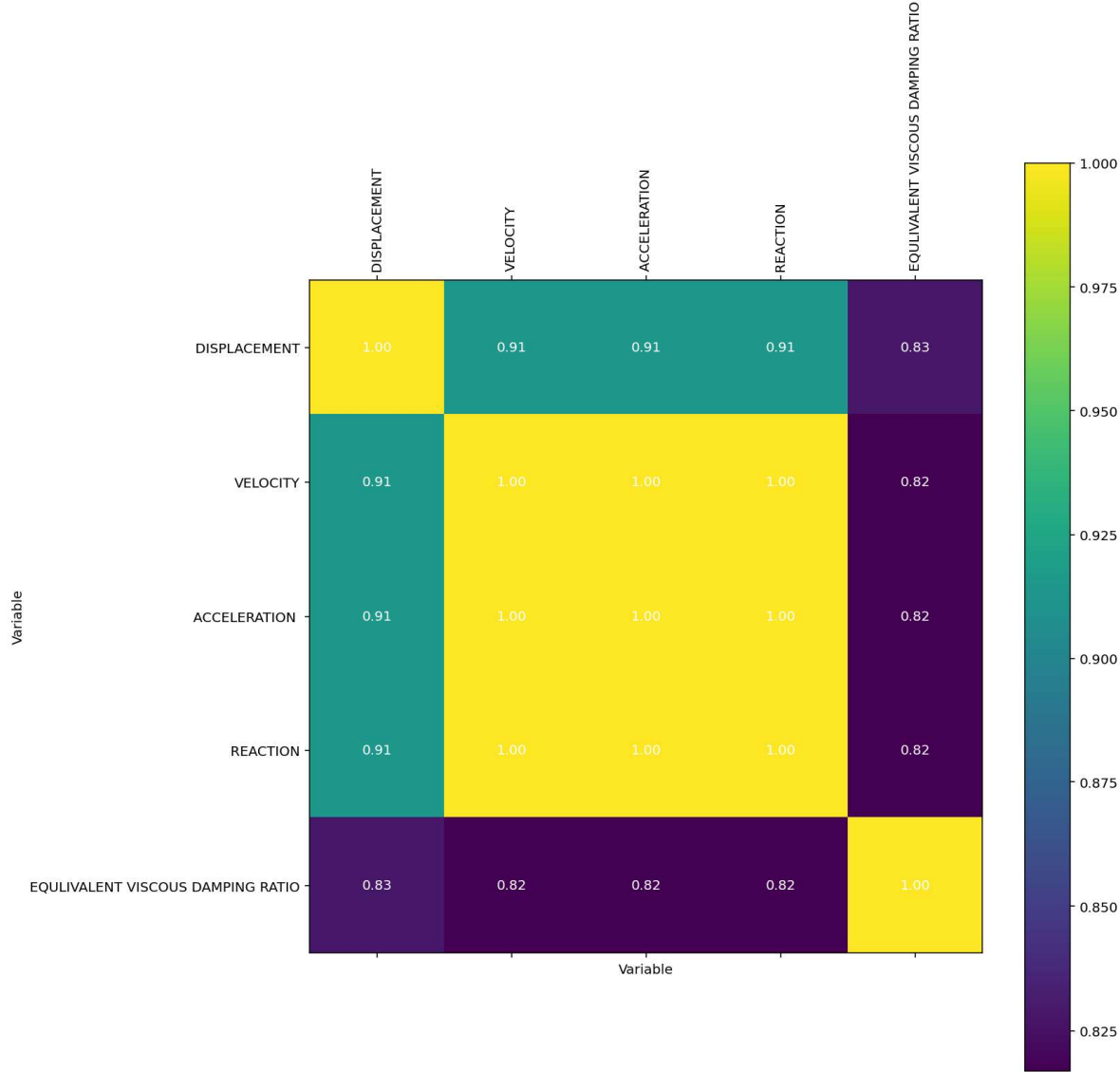


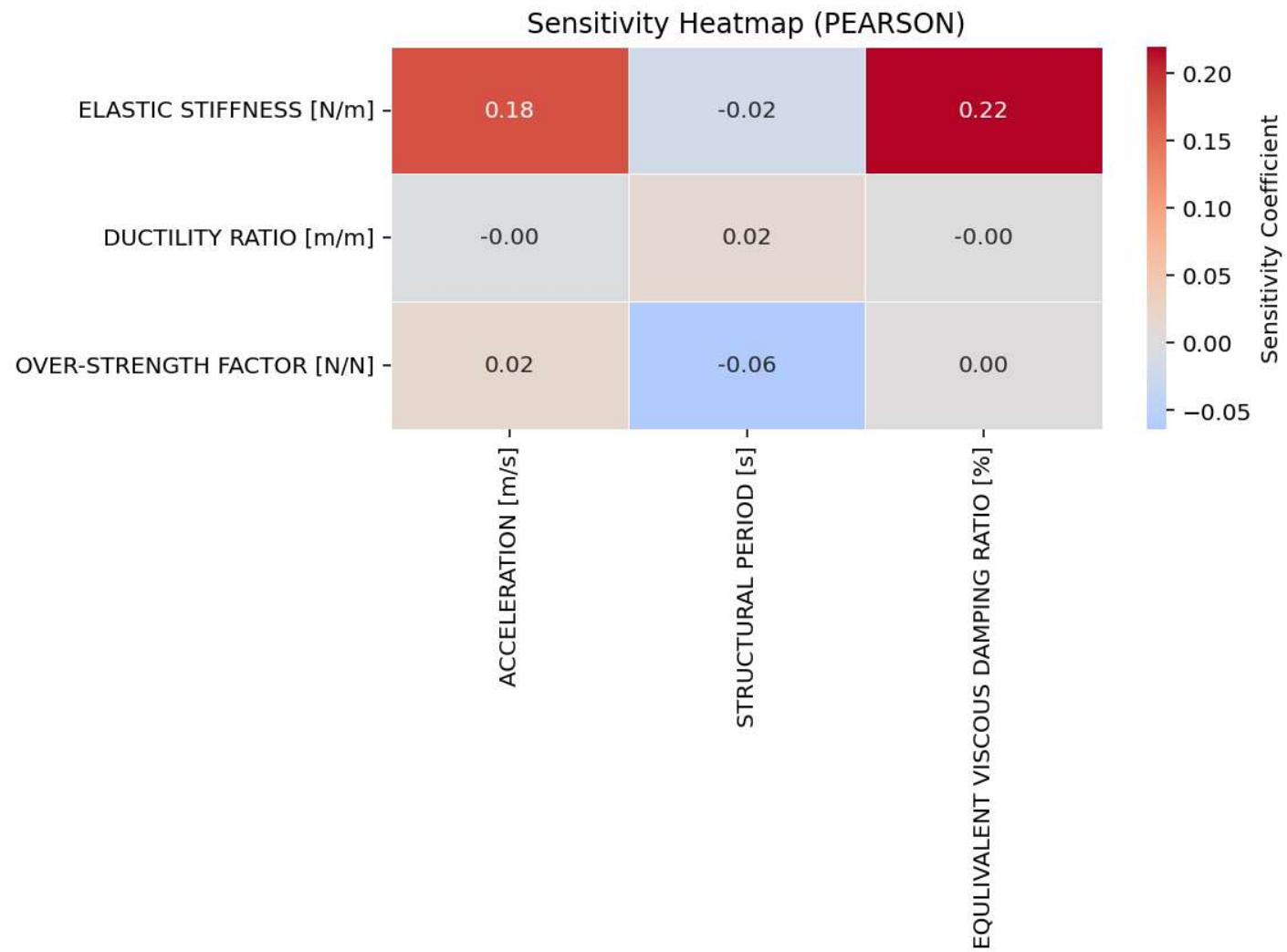
Histogram - Probability of Positive EQUIVALENT VISCOUS DAMPING RATIO [%] FROM FREE-VIBRATION ANALYSIS is 100.00 %



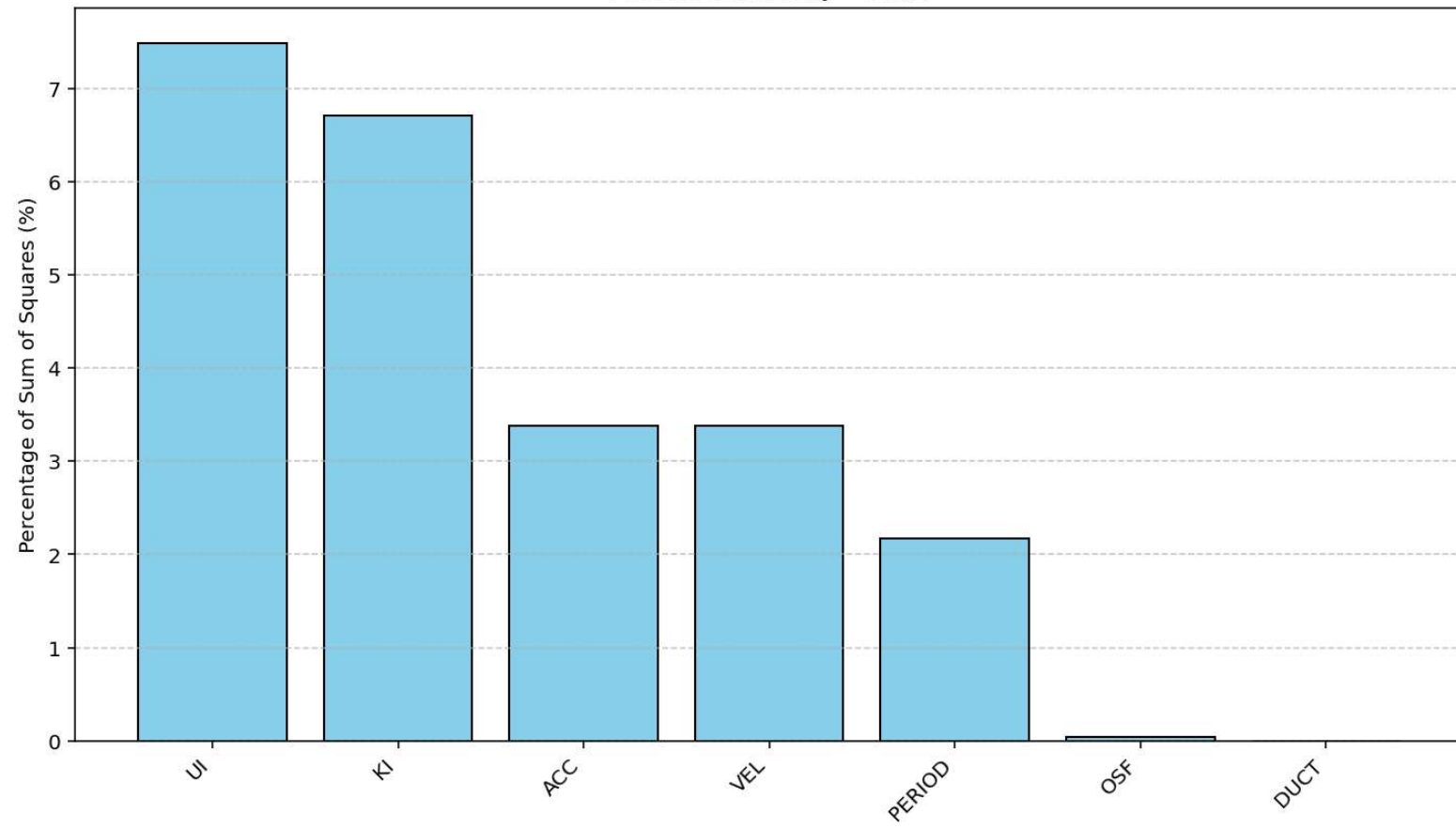


Correlation Heatmap





ANOVA Sensitivity - ZETA



ANOVA Sensitivity - ZETA

